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Structure, Properties and Bioelectric Potentials of Cancer Cells

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This paper examines cancer through the lens of biomedical engineering, focusing on the changes in the bioelectric properties of the cell membrane. While traditional oncology focuses on genetic factors, the emergence of “cancer bioelectricity” offers a new layer of information, in which the cell is treated as a dynamic electrical system.

The purpose of this paper is to investigate the relationship between cellular structure and the bioelectric signature of cancer cells. The analysis is organized around three central axes:

- i. **Scientific Basis (Theory)**, in which the structure–properties–bioelectric quantities (V_m , ionic fluxes, impedance, charges) are established,
- ii. **Technology Assessment**, in which techniques for measuring/mapping bioelectric signatures are compared, and
- iii. **Product Design**, in which a coherent framework for a device/sensor concept that exploits the bioelectric signature as a key input signal is proposed.

The paper concludes that bridging biological knowledge with engineering design can lead to innovative “bioelectronic medicine” (electroceuticals) tools for addressing the disease.

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Chapter 1 - Introduction

GLOBOCAN 2022 estimates show that cancer remains a major global health problem, with an increasing burden (19,976,499 new cases with 9,743,832 deaths in 2022) and a projected sharp increase by 2050 (53,504,187), mainly due to demographic ageing and inequalities in access to care.¹



Figure 1a: Global cancer burden (new cases, deaths and 5-year prevalence), GLOBOCAN 2022 estimates / Global Cancer Observatory (IARC/WHO).

At the level of biological interpretation, a useful “skeletal” framework is provided by the functional capabilities (known as *hallmarks*) that cancer cells acquire during progression toward malignancy; this helps the concepts of “structure–properties–bioelectric signals” fit together organically.²

Cancerous diseases constitute a heterogeneous group of pathological conditions characterized by disruption of normal tissue homeostasis and by the acquisition of capabilities such as uncontrolled proliferation, survival under adverse conditions, invasion of neighbouring tissues and, at advanced stages,

¹ Prevalence in epidemiology refers to the total number of people alive on a given date who have been diagnosed with a disease within the preceding five calendar years.

² This perspective is particularly useful in a biomedical engineering paper, because it directly links the cell’s micro-architecture with properties that can be measured, modelled and potentially exploited technologically.

metastasis. In simple terms, cancer is a group of diseases in which certain cells of the body “rebel,” proliferate without control, and have the power to invade and spread to other parts of the organism.

Despite advances in surgical, pharmacological and radiotherapeutic approaches, the complexity of the disease as well as its biological heterogeneity limit effectiveness and are accompanied by adverse effects that affect quality of life.

According to chapter 3 of the book *Comparative Oncology* [4], it is stated that:

“The peculiarities specific for each neoplasm result from heterogeneity, the presence of subclones whose unpredictable appearance and variation [...] make difficult tumor therapy. In addition to these peculiarities of tumor cells, the following should be considered: [...] the variable cell sensitivity to cytostatics, radiation, etc.”

At the cellular level, the differences between normal and cancer cells are not only molecular; they are also reflected in structure and in measurable biophysical properties. Classic morphological features of malignancy include nuclear abnormalities (changes in size/shape, abnormal chromatin, prominent nucleoli) and changes related to the nucleus/cytoplasm ratio [4]. At the same time, the reorganization of the cytoskeleton and of adhesion/motility mechanisms constitutes a key substrate for migratory and invasive phenotypes, which are critical for progression and metastasis [5].³

In recent years, intense research interest has developed around the role of bioelectricity in cancer, i.e. how electrical signals and charge distributions in cells and tissues relate to tumour initiation, growth and progression [6]. According to the scientific article “Membrane potential and cancer progression”:

³ This viewpoint is particularly useful in a biomedical engineering paper, because it directly links the cell's micro-architecture with properties that can be measured, modelled, and potentially exploited technologically.

“Emerging evidence suggests that the V_m [membrane potential] also plays important functional roles in non-excitable cells... changes in V_m , representing the long-term, slowly changing bioelectric gradient... have been shown to control critical cell functions including proliferation, migration, and differentiation.”

A key quantity is the membrane potential (V_m or V_{mem}), which arises from the asymmetric distribution of ions and from the selective permeability of the cell membrane via ion channels and transporters. In many cancer cell lines a depolarized V_m has been reported (less negative compared with corresponding healthy cells), a finding that is functionally linked to proliferation, to the regulation of cell volume, and to migration/infiltration mechanisms [6]. Furthermore, dysregulation of the expression and/or function of ion channels may contribute to many of the “functional capabilities” of cancer, a fact that has led to the framework of *oncochannelopathies*, in which ion channels are treated as critical regulators of cancer phenotypes. According to [6]:

“Given the increasing evidence showing that ion channels/transporters functionally participate in cancer progression, V_m might be a valuable clinical marker... and could even be artificially modified in order to inhibit tumor growth.”

The bioelectric differentiation of cancer cells is not limited to V_m . Studies using electrophysiological and electrical techniques (electrophysiology recordings, electrical impedance spectroscopy, etc.) indicate that these changes arise from the interaction of multiple factors and can be described as “bioelectric signatures.” According to [6]:

“Bioelectrical impedance analysis, which determines tissue electrical properties, has shown promise as a prognostic indicator to monitor cancer progression.”

For example, it has been reported that cancer cells may show higher membrane impedance (hence lower electrical conductivity), a change that may be correlated with behaviours such as increased proliferation, reduced apoptosis and increased directionality of migration/invasion. At the metabolic level, the Warburg effect⁴ may contribute to the export of anions (lactate) and to changes in surface charge, while at the level of cell “populations” it has been described that endogenous current/field gradients can emerge in mono- or multi-layers and tissues:

“Cancerous cells exhibit anomalies of both glycolysis and the tricarboxylic acid (TCA) cycle... tumor cells behave like a metabolic parasite.”

It is also important that bioelectric dysregulation does not concern the cancer cell in isolation, but is embedded in the tumour microenvironment and in cell-communication mechanisms, such as extracellular vesicles, which can participate in signal mediation in carcinogenesis (initiation–promotion–progression).

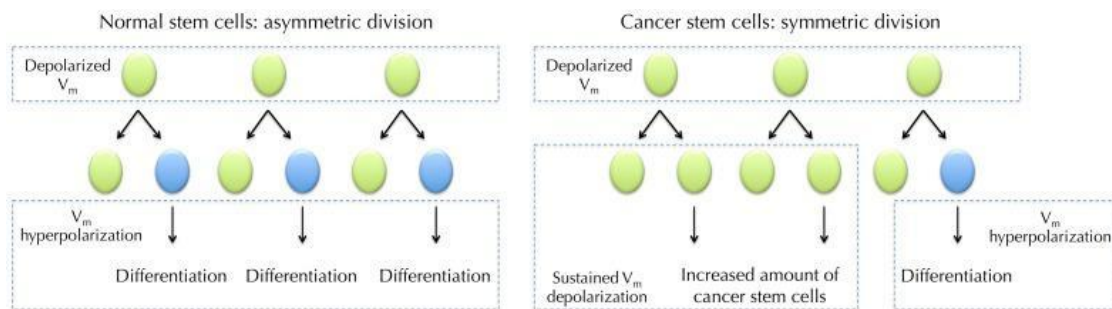


Figure 1b: V_m in normal stem cell (SC) differentiation and hypothesized role for V_m in cancer stem cells (CSCs).

The prospect of “cancer bioelectricity” is twofold: on the one hand it can enhance the fundamental understanding of cancer biology, and on the other it offers technologically exploitable signals for

⁴ The Warburg effect is the tendency of many cancer cells to produce energy mainly through glycolysis and to convert glucose into lactate, even when sufficient oxygen is available.

applications in diagnosis/prognosis and targeted intervention [6]. For example, approaches are being developed and/or clinically evaluated that use electric fields to affect critical cellular processes (mitosis) and/or ion-channel activity, seeking effectiveness with a different side-effect profile compared with classic systemic therapies. At the same time, the broader field of “bioelectronic medicine / electroceuticals” seeks to exploit electrical signals/stimulation for the therapeutic control of biological systems, pointing to a broader technological framework within which cancer, too, can be positioned as a target.

Chapter 2 - Scientific Basis / Theory

2.1. Chapter introduction

The study of cancer cells within the framework of Biomedical Engineering requires “stepping away” from traditional genetic analysis and moving towards a systemic, biophysical perspective. In this chapter, the cell is treated not simply as a biological entity, but as a dynamic, non-linear electrical system that is embedded within a broader multicellular network.⁵

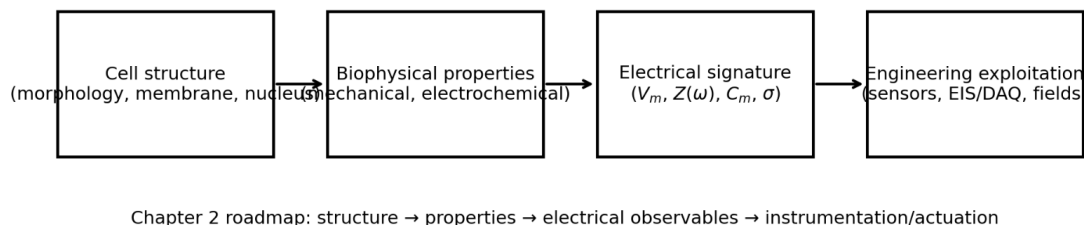


Figure 2.1: Roadmap of the chapter: from cell structure to biophysical properties, to electrical observables, and finally to engineering exploitation (sensors/fields).

The process of carcinogenesis is modelled as a fundamental failure of the feedback control of this network, accompanied by strictly measurable changes. Specifically, the structure, physicochemical properties and bioelectric potentials of the cell undergo radical changes that are reflected in electrical quantities such as complex impedance, membrane capacitance and electrical conductivity.

According to the scientific article “Nanotechnology and Cancer Bioelectricity” [7], the study of this bioelectric behaviour offers new, critical tools in the field of engineering:

“Cancer cells are known to exhibit abnormal electrical properties compared to their healthy counterparts, and this has driven researchers to investigate the potential of

⁵ This perspective of modelling the cancer cell as a dysfunctional electrical system is very important, as it allows the translation of medical pathology into measurable control and impedance quantities.

harnessing bioelectricity as a tool in cancer diagnosis, prognosis, and treatment.”

Understanding this theoretical background constitutes the “cornerstone” for the subsequent development of diagnostic sensors and therapeutic devices based on the analysis and manipulation of electric fields.

2.2. Structure of cancer cells

The architecture of the cancer cell differs significantly from that of a healthy cell, a fact that directly affects its electromagnetic and mechanical behaviour. Understanding this structure is a prerequisite for creating accurate equivalent circuits.

2.2.1. Morphological changes

At the macroscopic cellular level, the pleomorphism and irregular shape of cancer cells lead to a fundamental geometric change that can be quantified through the Fractal Dimension. While healthy cells present smooth outlines with a fractal dimension close to unity, the disorganized growth and increased cell density of cancerous tissues significantly increase their fractal dimension.

This geometric complexity affects the distribution of electrical charges on their surface and modifies the tissue’s ability to polarize under the influence of external electric fields, a phenomenon reflected in the dispersion index α during impedance spectroscopy (ideal capacitor, distribution of time constants).

In short: an increase in effective area due to folding leads to an increase in capacitance.

As also noted in a scientific study on the geometric analysis of tumours [8]:

“This study proved the capability of fractal dimension to distinguish cancer cells from healthy cells... it is clear that

the healthy cells have a smaller fractal dimension on average.”

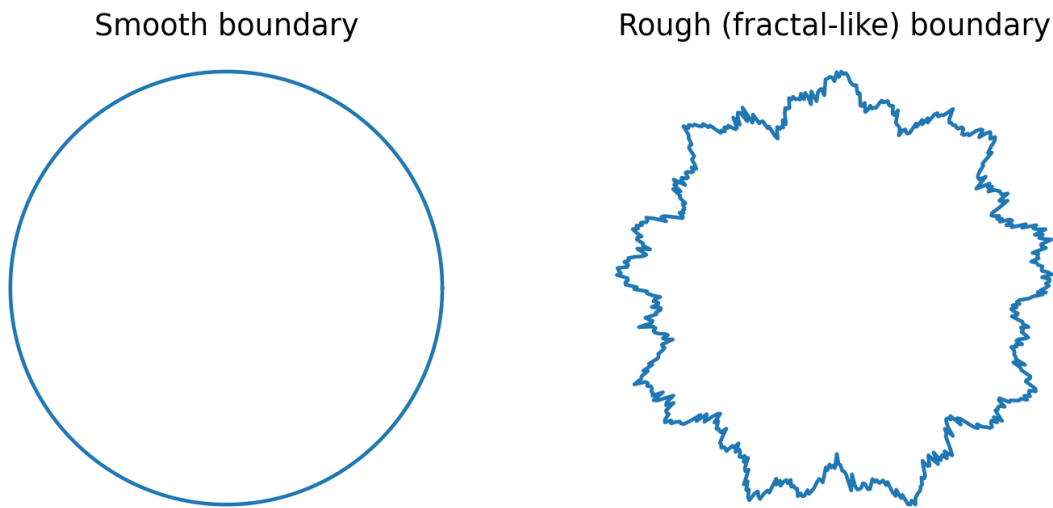


Figure 2.2a: Smooth outline (healthy) versus rough/“fractal-like” (cancerous), which increases the effective surface area and affects polarization/capacitance.

2.2.2. Cytoskeleton

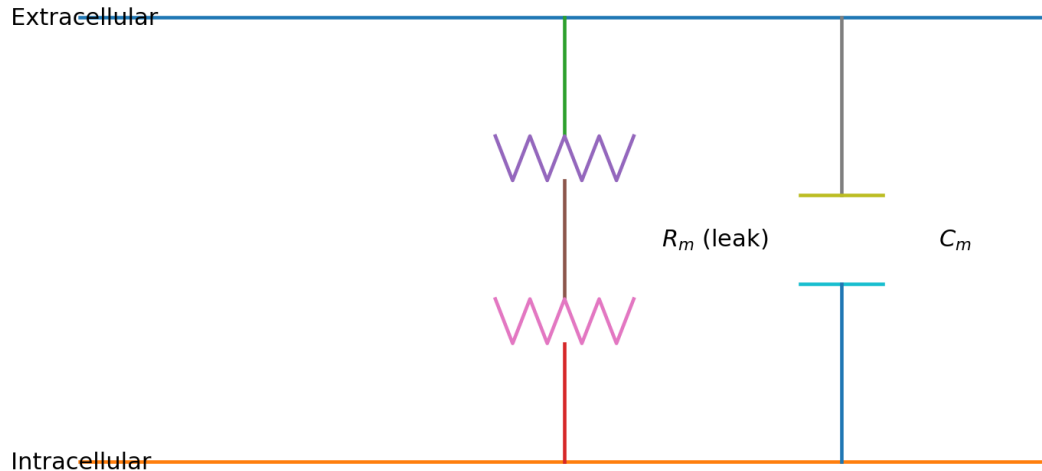
The cytoskeleton, composed of actin filaments and microtubules, functions as a dynamic, structural scaffold. In cancer cells, extensive disorganization of this scaffold is observed, a fact that confers on the cell increased motility and invasiveness.

From the perspective of biomechanics, this structural degradation turns the cell into a particularly deformable material. The loss of structural integrity allows the cell to manoeuvre through the extracellular matrix (ECM) and to enter the bloodstream, constituting the primary mechanical cause of metastasis.

2.2.3. Cell membrane

The cell membrane is the most critical interface boundary from the electrical engineer’s point of view, as it functions as an imperfect, thin dielectric insulator separating two conductive electrolytes (the intracellular and extracellular fluid).

Leakage/shunt behaviour is treated accordingly. This is consistent with the “leaky capacitor” model referred to explicitly in bioelectric-signal lecture notes (“Dialeksi-3b-4-Sensors_ch2.pdf”).



Cell membrane equivalent: parallel leakage resistance and membrane capacitance

Figure 2.2b: Leaky-capacitor model of the membrane ($R_m \parallel C_m$): the membrane equivalent as a “leaky capacitor” — the membrane behaving with ohmic leakage between two electrolytes.

In malignancy, changes are observed in phospholipid composition, differentiation of the glycocalyx and, above all, overexpression of specific ion channels. These physical alterations radically change the dielectric constant of the membrane and lead to an increase in the effective conductive surface, reducing the cell’s shunt (leakage) resistance.

2.2.4. Nuclear architecture

The increased nucleus-to-cytoplasm ratio (N/C ratio) and the disorganization of chromatin are not merely indicators of cell

division, but also modify the total volume of organelles within the cytoplasm.

Because the nucleus has a different density (typically $>1200 \text{ kg/m}^3$) and dielectric behaviour compared with the cytoplasm, its enlargement changes the dispersion parameters in high-frequency measurements, affecting the way current passes through the internal volume of the cell.

2.3. Properties of cancer cells

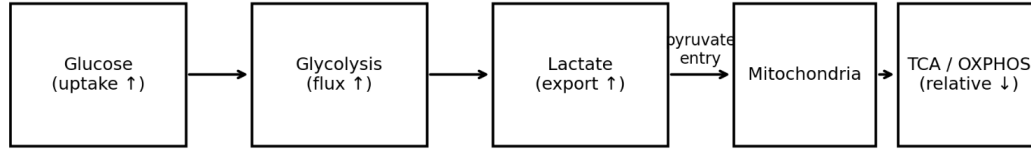
The structural changes described above translate into a series of macroscopic properties which constitute the input variables for sensor design.

2.3.1. Biological properties

Uncontrolled proliferation, resistance to apoptosis, and metastatic potential arise from the reprogramming of the cell. In systems theory, this is described as the transition of the biological network to a new, pathological “attractor” within the state space. The cancer cell ceases to function as part of a cooperative whole and behaves as an isolated, single-cell organism whose control function focuses exclusively on its own survival.

2.3.2. Physicochemical properties

The Warburg effect, i.e. the production of energy mainly through glycolysis even in the presence of oxygen, makes the cancer cell a “metabolic parasite.” The continuous secretion of lactic acid creates an acidified microenvironment (low pH) around the tumour.



Warburg phenotype: aerobic glycolysis with lactate production

Figure 2.3a: Warburg phenotype (physicochemical properties): increased glycolysis and lactate production/export, with effects on the microenvironment (pH/ionic balance).

The increased concentration of ions in this extracellular space leads to a measurable increase in the local electrical conductivity of the tissue, a change that is exploited by bioimpedance tomography techniques for locating the mass.

As also reported by the relevant literature [9] (Baba AI, Cătoi C. *Comparative Oncology* (2007), Chapter 3: “Tumor Cell Morphology” (NCBI Bookshelf)):

“Cancerous cells exhibit anomalies of both glycolysis and the tricarboxylic acid (TCA) cycle... tumor cells behave like a metabolic parasite.”

2.3.3. Mechanical properties

Through the use of nano-indentation techniques, such as Atomic Force Microscopy (AFM), it has been demonstrated quantitatively that individual cancer cells are significantly softer than healthy ones.

For example, Young’s elastic modulus (E) in healthy breast cells (MCF-10A) ranges from 0.2 to 0.9 kPa, while in the corresponding cancerous cells (MCF-7) it shows a drop, ranging from 0.1 to 0.4 kPa.

Table 1. Comparison of mechanical properties between cancerous and normal cells [T1]

	Normal cell		Cancer cell	
	Cell line	Young's modulus, E (kPa)	Cell line	Young's modulus, E (kPa)
Breast cancer	184A	2.26 ± 0.56	T47D, MCF-7	1.20 ± 0.28 , 1.24 ± 0.46
	MCF-10A	0.2–0.8, 0.5–0.9	MCF-7	0.1–0.4

Table 1: Comparison of mechanical properties between cancerous and normal cells.

Kelvin–Voigt (spring || dashpot)

Maxwell (spring — dashpot)

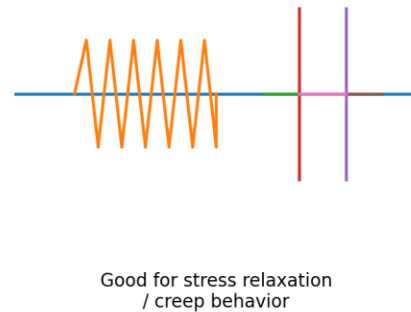
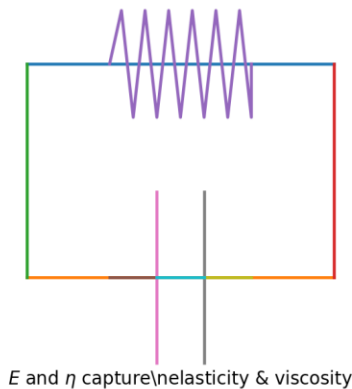


Figure 2.3b: Rheological equivalents (Kelvin–Voigt/Maxwell) for modelling the cell as a viscoelastic material, with parameters that are often reduced in cancer cells.

For a complete description of these properties, simple Hookean elasticity is not sufficient. Cells are simulated using advanced rheological models (such as the generalized Kelvin–Voigt model or the fractional Maxwell model), incorporating spring–dashpot elements to describe the viscoelastic relaxation and deformability required for metastasis.

According to a relevant scientific publication [10]:

“A general view is that single cancer cells are more deformable and their viscoelastic parameters are reduced as compared to their normal counterparts... This has been shown for different types of cancers including bladder, prostate, cervix, ovary and breast cancer.”

2.3.4. Electrical properties

The most exploitable aspect for diagnosis is the electrical signature of cancer. Through Electrical Impedance Spectroscopy (EIS), the cell's behaviour with respect to alternating current is analysed. The increase in folds and microvilli on the cell membrane causes a rapid increase in the Specific Membrane Capacitance (C_m).

Experimental data indicate that highly metastatic cells (such as MDA-MB-231) show more than double the capacitance (approximately 22.3 mF/m^2) compared with normal cells ($\sim 8\text{-}9 \text{ mF/m}^2$).

In addition, the intracellular fluid of cancer cells shows significantly increased conductivity at frequencies on the order of MHz. When cancerous tissue is analysed as a whole, the Cole-Cole model reveals deviations in the relaxation frequency (f_c) and reduced complex impedance, offering a robust, non-invasive biomarker.

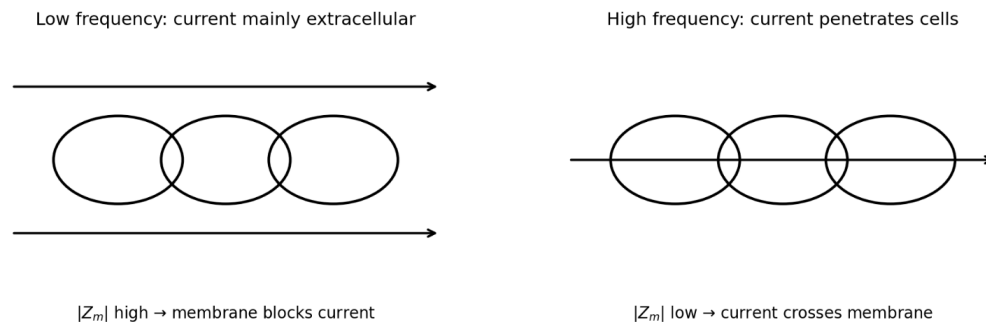
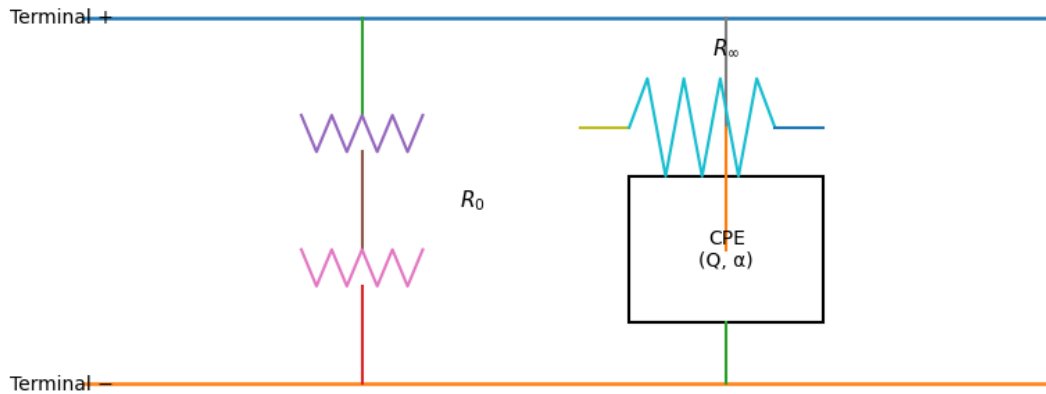


Figure 2.3c: Current flow at low vs. high frequencies (intuitive EIS): at low frequency the membrane “blocks” current (mainly extracellular flow), while at high frequency the capacitive reactance decreases and current passes through the cells.

As the literature of the field characteristically concludes [11]:

“Cancer cells of different cell origin possess their own signature electrical parameters, especially when compared with their normal counterparts... This approach paves the way for exploring unique electrical signatures to enable their detection and discrimination.”



Cole-Cole / Cole element: R_0 in parallel with (R_∞ in series with CPE)

Figure 2.3d: Cole-Cole equivalent circuit. The extracellular resistance (R_e) is shown in parallel connection with a branch comprising the intracellular resistance (R_i) and a Constant Phase Element (CPE), which models the non-ideal capacitance of the tumour cell membrane.

2.4. Bioelectric potentials of cancer cells

The bioelectric signature extends beyond passive electrical properties, incorporating the cell's active potentials. The cell functions as a microscopic battery, and the regulation of its voltage constitutes a critical control mechanism.

2.4.1. Membrane Potential (V_m)

At rest, healthy, non-excitabile cells maintain a potential difference (membrane potential) that is strictly confined to the range: $V_m \approx -60$ to -90 mV.

In contrast, a fundamental characteristic of cancer cells is their permanent depolarization, with the potential shifting to values between: $V_m \approx -10$ to -50 mV.

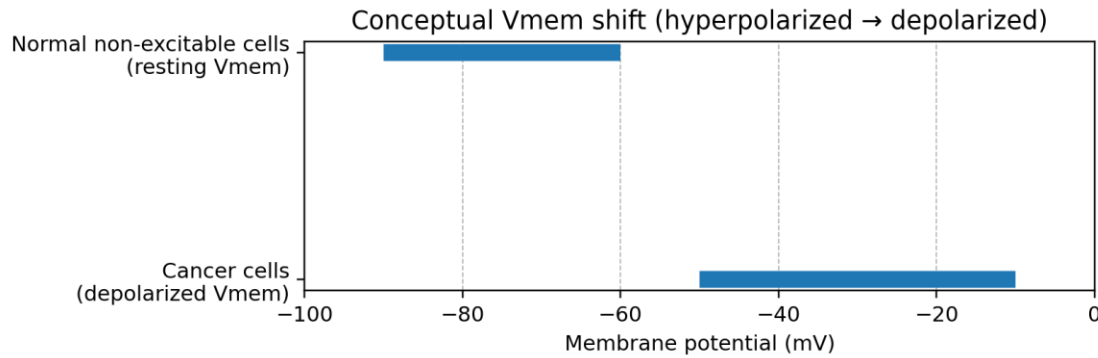


Figure 2.4a: Conceptual shift of the membrane potential towards depolarization in cancer cells, as a measurable bioelectric signal (biomarker).

This drop in the “operating voltage” is not merely a symptom, but a powerful instructive cue. Depolarization bypasses normal constraints and instructs the cell to initiate the cell cycle, to proliferate continuously, and to adopt migratory behaviour (it regulates Ca^{2+} signalling).

According to the scientific work of Chernet and Levin [12]:

“Abnormal depolarization of resting potential (V_{mem}) is a convenient marker for neoplasia and activates a metastatic phenotype in genetically-normal cells in vivo.”

2.4.2. Ion Channels (Oncochannelopathies)

The membrane potential is controlled by pumps and ion channels. In electrical circuit theory (based on the Hodgkin–Huxley model), these channels are modelled as non-linear, voltage-dependent specific conductances (g) connected in series with voltage sources representing the Nernst potential of each ion. In cancer, the dysfunction and overexpression of these channels constitute a primary cause of pathology, establishing the term “oncochannelopathies.” These particular electrical gates do not merely transport charge, but alter the circuit’s time constants, keeping the cell permanently in the depolarized state required for invasion.

As also pointed out in the relevant literature [13]:

“Given the increasing evidence showing that ion channels/transporters functionally participate in cancer progression, V_m might be a valuable clinical marker... and could even be artificially modified in order to inhibit tumor growth.”

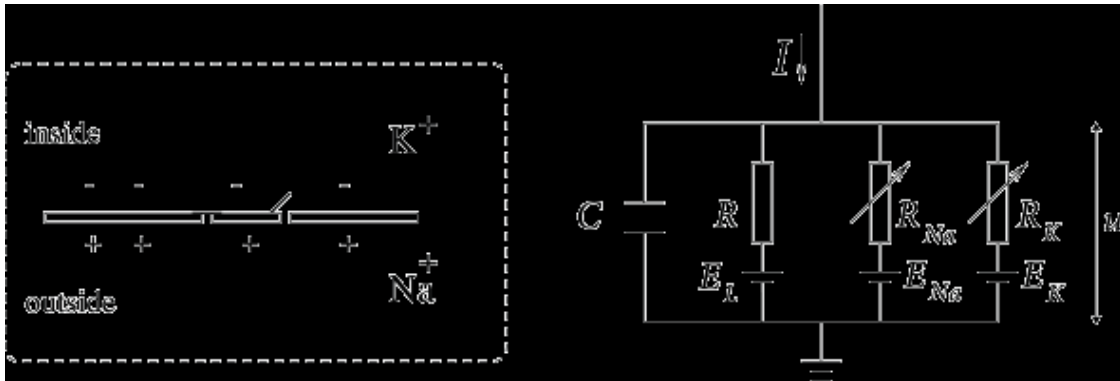
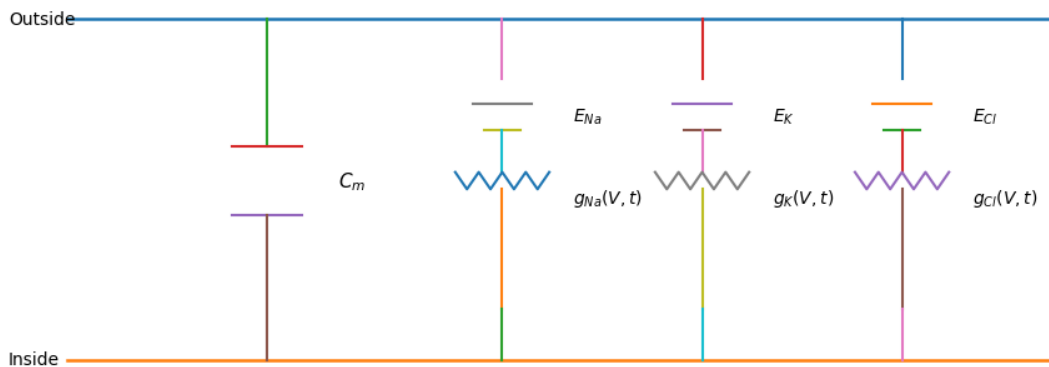


Figure 2.4b: Hodgkin–Huxley electrical model.



Membrane model: C_m in parallel with ionic branches (conductances + Nernst batteries)

Figure 2.4c: The cell membrane (C_m) is placed in parallel with the branches for sodium (Na), potassium (K) and chloride (Cl) ions. Each branch contains a Nernst-potential battery (E) and a variable resistance (R) determined by the action of the ion channels.

2.4.3. Role of Ca^{2+}

Ca^{2+} (calcium) functions within the cell as a critical second messenger. Changes in membrane voltage control the opening of voltage-gated calcium channels.

In engineering terms, the calcium current acts as the trigger signal of an automation system that controls the cytoskeleton, the contractions used for cell movement, and the deactivation of the self-destruction process (apoptosis).

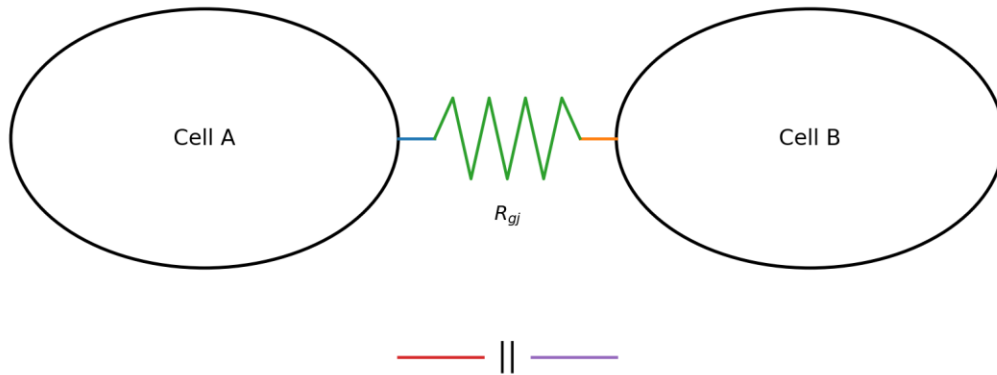
2.4.4. Electrical coupling

The successful homeostasis of a tissue is based on uninterrupted intercellular communication. This is ensured through gap junctions, which function as conductive connections of variable resistance between the cytoplasm of neighbouring cells. Through these, the entire tissue forms a coherent electrical network (syncytium).

Gap-junction conductance changes dynamically, thereby affecting the time constants of the network.

One of the first steps of carcinogenesis is the increase in ohmic resistance or the complete severing of these connections. The cancer cell creates an “open circuit” with respect to the surrounding tissue, a fact that allows it to maintain its own pathological potential (V_m) without being “grounded” or regulated by the electrical synchronization of the rest of the healthy organism.

Healthy tissue: electrical coupling via gap junction



Cancer phenotype: reduced/absent coupling ("open circuit")

Gap junctional intercellular communication (GJIC) ↓

Figure 2.4d: Gap-junction coupling versus “open circuit.”

This change has been documented historically, as noted in a relevant article [14]:

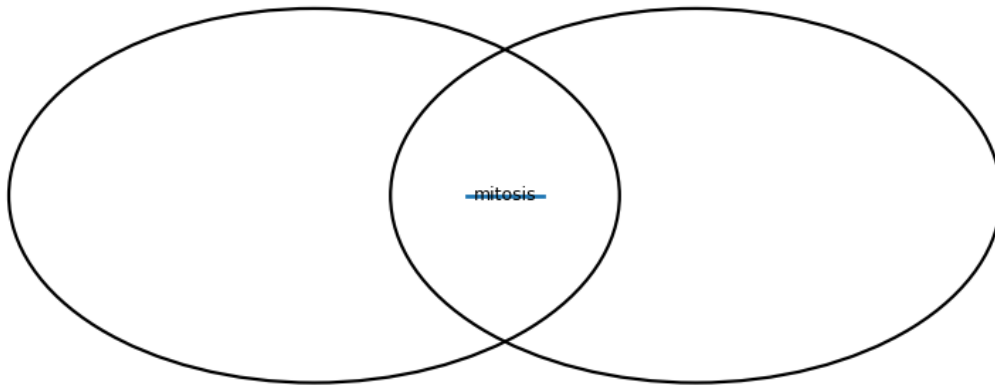
“Fifty years ago, tumour cells were found to lack electrical coupling, leading to the hypothesis that loss of direct intercellular communication is commonly associated with cancer onset and progression.”

2.4.5. Clinical significance

Recognition of the bioelectric quantities described above opens the way for purely electrical diagnostic and therapeutic methods. Tumor Treating Fields (TTFields), for example, exploit external alternating electric fields at specific frequencies (TTFields are typically described as intermediate-frequency, ~100–300 kHz, low-intensity, ~1–3 V/cm) that target the unique capacitive properties of cancer cells in order to disrupt the mitotic spindle during mitosis.

Figure 2.4e: Conceptual illustration of TTFields — an alternating, intermediate-frequency electric field interacting with polarizable structures during mitosis.

Alternating electric field (TTFields) ~100–300 kHz



Concept: AC fields interact with polarizable structures during cell division

Figure 2.4e: Conceptual illustration of TTFields.

Correspondingly, sophisticated Data Acquisition systems for measuring microscopic impedance offer indications of disease development in the blood long before macroscopic symptoms appear.

In this new era (bioelectronic medicine), the cancer cell is no longer addressed only with chemical drugs, but is modelled, measured, and ultimately “corrected” as a dysregulated circuit.

2.5. Multiscale bioelectric signature of cancer

The bioelectricity of cancer is not a single quantity (e.g. only V_m or only impedance), but a set of interconnected electrical parameters that manifest at multiple scales: at the level of the individual cell, at the level of populations/cohesive groups of cells

(cohorts, mono-/multi-layers), and ultimately at the tissue level (Single Cell → Cohorts → Tissues).

This perspective holds that the “bioelectric signature” must be treated as the state vector of an electrical system, one that is strongly influenced by the microenvironment (pH, ionic strength, ECM, mechanical cues), rather than as a static “biomarker” consisting of a single parameter.

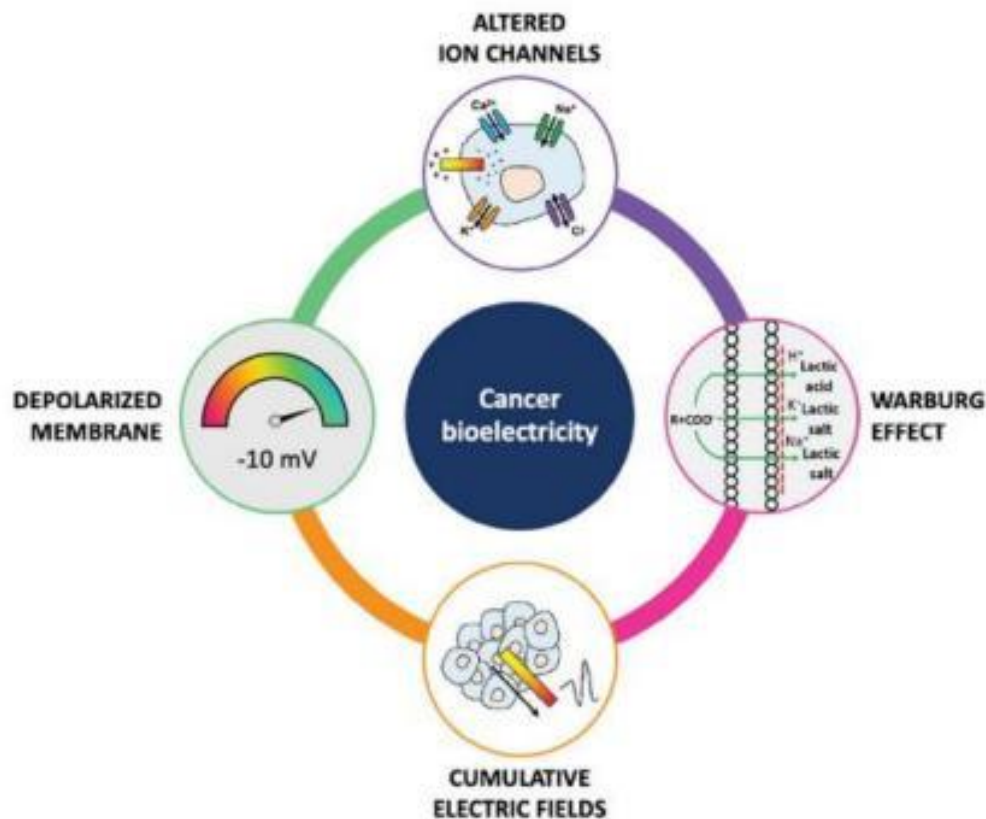


Figure 2.5a: Overview of the multifactorial interplay in cancer bioelectricity: altered ion channels → depolarization → Warburg effect/interface charge → endogenous electric fields.

A. Altered ion channels (changes in $\text{Na}^+/\text{K}^+/\text{Ca}^{2+}/\text{Cl}^-$ channels, etc.) B. Depolarized membrane (more depolarized V_m — in the figure ~ -10 mV) C. Warburg effect (metabolic reprogramming → export of lactate/ions, changes at the membrane) D. Cumulative electric fields (endogenous electric fields/currents in groups of cells/tissues)

2.5.1. The three components of the bioelectric signature

According to Moreddu's review ("Nanotechnology and Cancer Bioelectricity: Bridging the Gap Between Biology and Translational Medicine") [15], the bioelectricity of cancer cells can be summarized in three basic, interconnected components:

- (i) ionic currents,
- (ii) faradaic currents,
- (iii) surface charges/double layer.

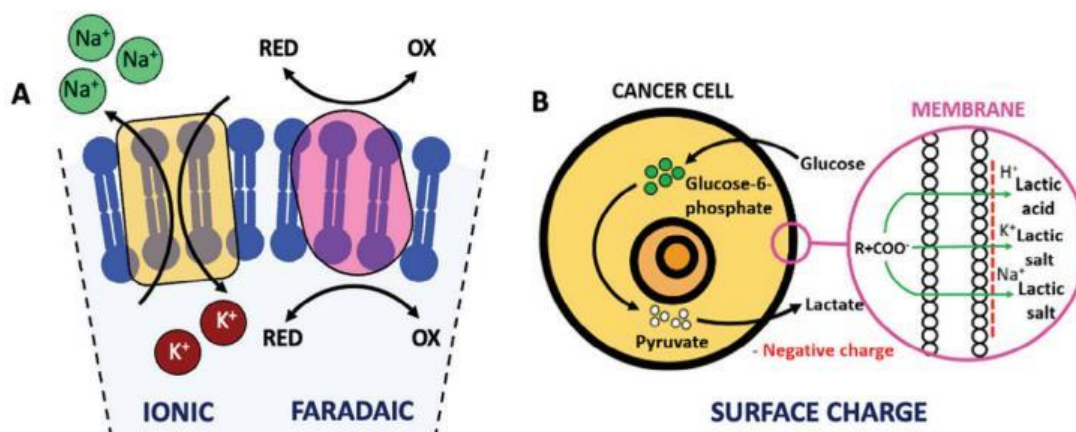


Figure 2.5b: Bioelectrical phenomena observed in single cancer cells. A) Ionic currents and faradaic currents; B) Warburg effect in cancer cells: negatively charged surface profile.

The bioelectric signature in single cells as the sum of ionic currents, faradaic (redox) currents, and surface charge (double layer).

- **Ionic currents:** determined by pumps/channels and shift the operating point of the “membrane circuit” (hence also V_m).
- **Faradaic (redox) currents:** arise from electron transfer in oxidation–reduction reactions (the electrochemical component of the signal).
- **Surface charges:** function as a boundary condition at the membrane/electrolyte interface (e.g. via negative net charge), affecting local fields and binding/interaction behaviours.

In the same context, the Warburg phenotype can contribute to a more negative surface charge due to the export of anions (e.g. lactate), so the “bioelectric signature” is not only channel electrophysiology but also an electrochemical interfacial signature.

2.5.2. From the individual cell to cohorts/tissues: endogenous fields and directed migration

Changes at the single-cell level can “scale up” and manifest as endogenous current gradients/electric fields in monolayers, multilayers and tissues (cohort effects).

Therapeutically, the reasoning is that electric fields can “tamper with” the behaviour of cancer cells, e.g. via ion-channel activity, targeting selective apoptosis, with the aim of fewer side effects than chemotherapy/radiation.

In practice, this means that cancer bioelectricity may not be described only as changes in the parameters of a single cell, but also as a “field distribution problem” in space, linked to cell communication, changes in adhesion, and ultimately migration/invasion phenomena.

2.5.3. The “engineering gap”: why measuring the signature is difficult in non-excitable (cancer) cells

Within the framework of the article [15], the core problem is the mismatch between signal and tools. Cellular electrical recording historically developed around action potentials in excitable cells (“cellular electrical recording mainly targeted... action potentials in excitable cells”), with microelectrode arrays as a classic example: “record extracellular signals in excitable cells... operating at high resolutions.”

In contrast, cancer/non-excitable cells often exhibit slow/static signals, “orders of magnitude smaller than action potentials,” and therefore require low input noise, high SNR at low frequencies, and long-term stability. Furthermore, at the cohort level, “electrical

activity” may lie in the range of 0.1–100 pA currents and 1–10 Hz frequencies, which makes commercial platforms (designed for neuronal signals) difficult to translate without redesign.

The promise is purely an engineering one:

“By customizing the electrodes and the readout systems... [devices] would be able to record much tinier electrical signals in non-excitabile cells...”

That is, optimizing the cell–electrode interface (reducing impedance, increasing effective area, an appropriate front-end/readout) so as to reach the pA/Hz range.

In the same context, the review emphasizes that nanotechnology can act as a “bridge” from biological findings to applications:

“nanotechnology may have the potential of bridging the gap between biology and translational medicine.”

Finally, the role of nano-bio tools is summarized in two complementary directions:

1. **Scientific advancement:** “Bespoke nanosystems may allow investigating cancer cells electricity... in finer details and higher throughput...,” for mapping/modelling signatures per subtype/stage.
2. **Technological advancement (translation):** nanosystems that “detect cancer cells electricity with high accuracy and specificity” and even “reprogram bioelectricity at the nanoscale...,” i.e. moving from measurement to intervention.

2.6. Cancer reversion therapy

Cancer reversion therapy describes a strategy that does not primarily aim at eliminating cancer cells, but at shifting the cellular state towards a less malignant, more differentiated and/or functionally “normalized” state. In its strict sense, this means stable and heritable phenotypic normalization, which is maintained

even after the intervention is stopped, and is accompanied by a substantial reduction in tumorigenic capacity. [2,6,1]

2.6.1. Summary

Cancer reversion therapy describes a strategy that does not primarily aim at eliminating cancer cells, but at shifting the cellular state towards a less malignant, more differentiated and/or functionally “normalized” state. In its strict sense, this means stable and heritable phenotypic normalization, which is maintained even after the intervention is stopped, and is accompanied by a substantial reduction in tumorigenic capacity. [2,6,1]

“Reprogramming of cancer cells” is a broader framework, which includes:

- (a) restoration to more “normal” attractor states,
- (b) induction of differentiation (differentiation therapy),
- (c) redirection of lineage/transdifferentiation (transdifferentiation/lineage conversion),
- (d) epigenetic or transcriptional re-loosening of plasticity, as well as
- (e) “reprogramming” of the microenvironment (TME) that stabilizes or destabilizes malignant states. [2,6,2]

The clinical applicability of these strategies depends strongly on the type of cancer: in haematological malignancies, and particularly in acute promyelocytic leukaemia, there is already strong evidence of effectiveness, whereas in most solid tumours stable reversion remains more difficult due to heterogeneity and plasticity. [2,6,3]

The “best documented” clinical form of reversion/reprogramming is considered to be differentiation therapies:

- (i) ATRA + arsenic trioxide in acute promyelocytic leukaemia, with excellent results in a randomized phase III trial,

- (ii) IDH inhibitors in AML, which “unblock” differentiation, with characteristic toxicities of the differentiation-syndrome type, and
- (iii) retinoids as differentiation/maintenance therapy in selected paediatric neoplasms (e.g. neuroblastoma). [2,6,4]

The most well-documented forms include differentiation therapies, such as ATRA + arsenic trioxide, IDH inhibitors, and certain retinoids. Despite their theoretical and clinical importance, critical issues remain: the precise definition of “true” reversion, the risk of partial reprogramming, the safety of targeted interventions, and the possibility of long-term disease stabilization. [2,6,5]

The core idea is that, instead of “switching off” the system, we attempt to move it from a pathological stable state to a more physiologically normal operating state.

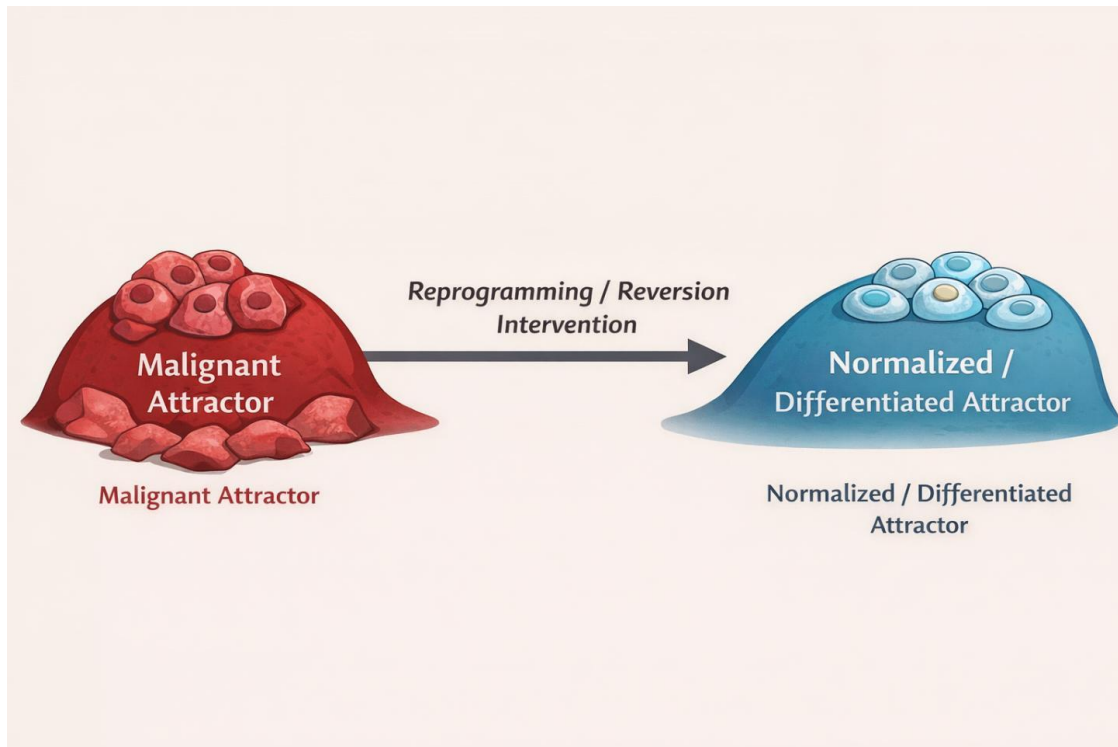


Figure 2.6.1: Schematic depiction of cancer reversion therapy as the shift of the cancer cell from a stable malignant state towards a more differentiated and functionally normalized state.

2.6.2. Definitions, terminology, and historical context

The concept of phenotypic reversion/reversal of cancer has historical roots as far back as the 1950s, and describes cases in which cells with malignant characteristics are led, under appropriate conditions, to a more organized and less aggressive state. The field was reinvigorated in the modern era thanks to advances in cellular reprogramming, epigenetics, and single-cell analysis, which showed that cellular identity is not absolutely fixed but can be reshaped. [2,6,6]

A central distinction is that between stable reversion, in which the new phenotype is maintained even after the stimulus is removed, and transient, stimulus-dependent plasticity, in which “normalization” is lost as soon as the intervention is withdrawn. [2,6,7] At the same time, true reversion must be distinguished from phenomena such as simple cell-cycle arrest, dormancy, or

senescence, which may be therapeutically useful but do not equate to a return to a non-malignant state. [2,6,8][2,6,9]

Historically, important experiments in embryonic microenvironments and blastocysts showed that the microenvironment can redirect phenotype, demonstrating that genotype alone does not fully determine cell behaviour. [2,6,10]

The modern link to reprogramming cancer cells was strengthened by the discovery of iPSCs, which demonstrated that cellular identity is reversible through transcriptional and chromatin networks, although limitations remain, such as partial reprogramming, safety, and difficulty in interpreting biological outcomes. [2,6,11]

The core idea is that it is not enough to see whether the cell “changed,” but whether the new state is stable without continuous external stimulation — that is, whether the system has truly moved to a new equilibrium point.

2.6.3. Biological mechanisms and molecular reprogramming networks

The biological mechanisms of cancer reversion therapy become more understandable if cancer is viewed as a stabilized cellular state that results from the combined action of genetic alterations, epigenetic changes, and signals from the microenvironment. [2,6,12] Within the framework of systems biology, this means that cancer cells are “trapped” in malignant attractor states, and the therapeutic goal is to move them to more functional, less malignant states, while simultaneously manipulating both intracellular and extracellular parameters so as to prevent relapse. [2,6,13]

At the molecular level, reversion can be pursued along 4 axes:

1. **Epigenetic remodelling**, where changes in DNA methylation, histones, and chromatin affect plasticity and resistance; [2,6,14]

2. **Transcriptional networks**, where transcription factors and chromatin regulators control cell-identity programmes; [2,6,15]
3. **Intracellular signalling**, via pathways such as MAPK/ERK, PI3K/AKT/mTOR, TGF- β , WNT/ β -catenin and Notch, which regulate transitions between stemness, differentiation, and EMT/MET; [2,6,16] and
4. **Microenvironment/tissue architecture**, where the ECM, mechanical tension, and cell–cell interactions can stabilize or destabilize the malignant phenotype. [2,6,8]

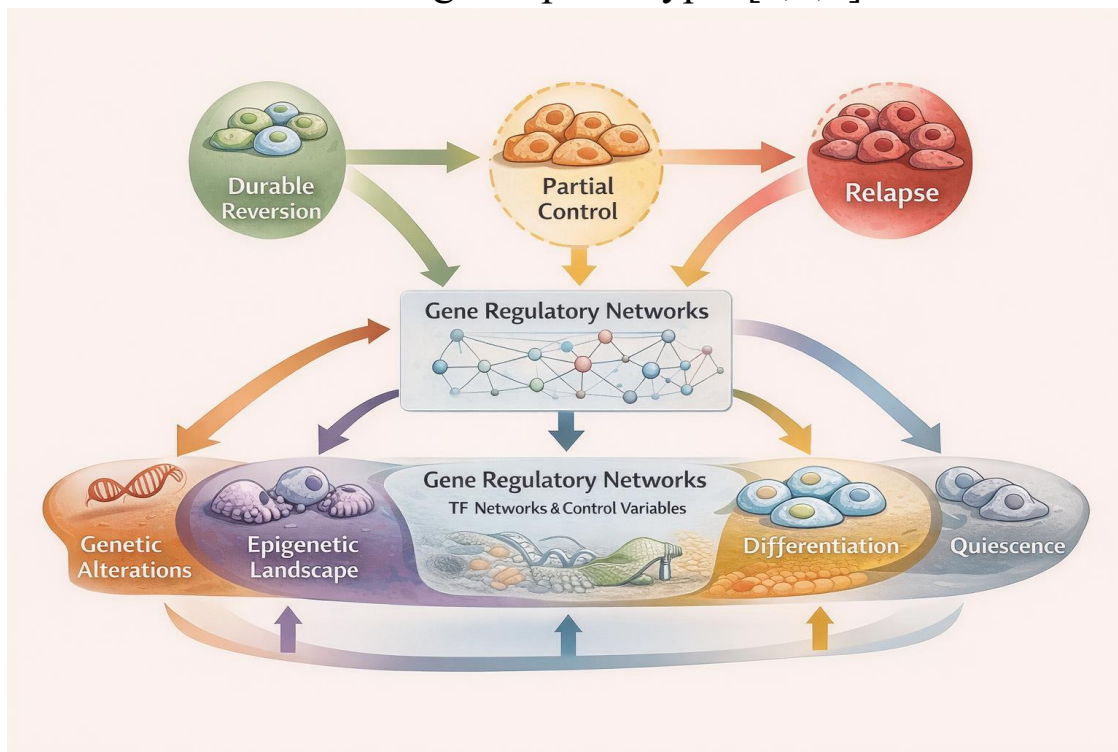


Figure 2.6.2: Multi-level control model of cellular state: genetic, epigenetic, and microenvironmental inputs shape the regulatory networks and determine the transition between malignant and non-malignant attractor states.

It is important that reversibility does not always equate to stable reversion. There are cases where temporary “normalization” is achieved — for example in 3D cultures with β 1-integrin inhibition — but the phenomenon is lost when the stimulus is removed, so it is more a case of context-dependent regulation than a permanent change of identity. Correspondingly, subpopulations with

reversible drug resistance show that plasticity does exist, but it can also lead to persistent resistant states rather than true therapeutic reversion.

Seen from an electrical engineer's perspective, this resembles a multi-stable-state system: it is not enough to apply a momentary stimulus and observe a change in output; the control network itself and the boundary conditions must be changed so that the cell settles into a new, stable operating point.

2.6.4. Therapeutic and experimental platforms for “reversion”

Therapeutic and experimental platforms for cancer reversion can be classified according to the level of control at which they intervene. The most mature category is small molecules, which include classic differentiation agents such as retinoids and arsenic trioxide, targeted inhibitors such as IDH inhibitors, as well as epigenetic modifiers that remodel chromatin and transcriptional programmes. [2,6,17] More advanced but still translationally difficult approaches include gene editing and, in particular, epigenome editing, where tools such as dCas9-conjugated modifiers target the selective reactivation of tumour-suppressor networks without permanent DNA breaks. [2,6,18]

A third category is RNA therapeutics (miRNA mimics, anti-miR, siRNA), which target the “rewiring” of regulatory networks. Although this approach is theoretically powerful, the experience with MRX34 (miR-34a) showed that biological rationale alone is not sufficient, as immunotoxicity and the delivery vehicle can be critical limitations. At the same time, differentiation and transdifferentiation strategies seek either to push cancer cells towards terminal maturation or to move them into a different lineage identity, as has been shown experimentally in breast and glioblastoma models, although in solid tumours these applications remain mostly at an early stage.

Finally, reprogramming of the tumour microenvironment (TME) also plays an important role, through immunomodulatory interventions such as macrophage repolarization, STING activation, or CD40 agonists. Although these approaches do not directly reprogram the cancer cells themselves, they act as a critical extracellular arm for stabilizing the desired anti-cancer state. Overall, the most well-established clinical field remains differentiation therapies in haematological malignancies, while the other platforms range from translational to early experimental stage, with delivery, off-target activity, and safety as the main limitations.

From a technical standpoint, these platforms represent different “control layers”: some tools act at the biochemical level, others at the genomic/epigenetic level, and others at the boundary conditions of the system (microenvironment). The difficulty is not only in changing the state, but in making it stable and safe.

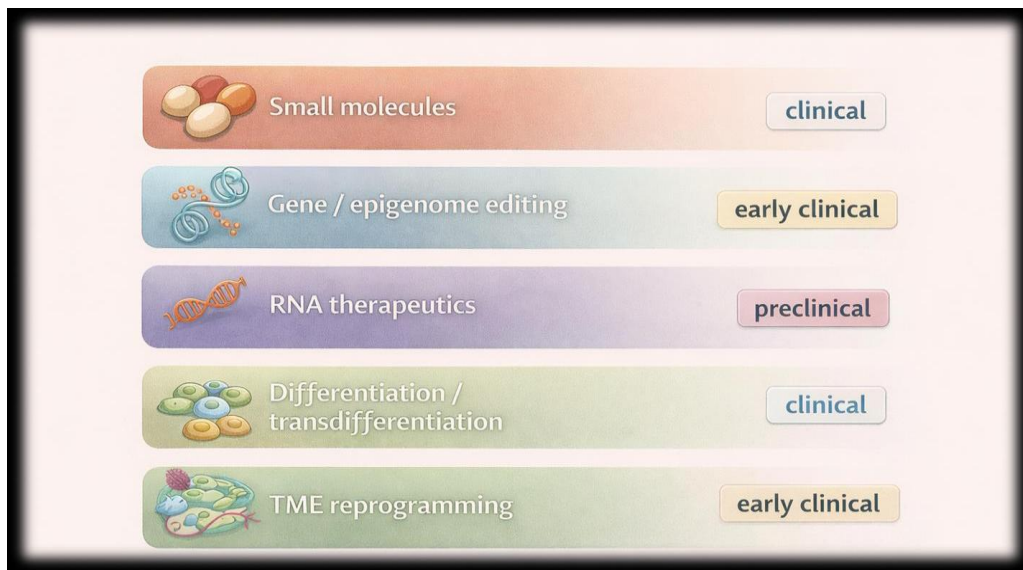


Figure 2.6.3: Classification of the main cancer-reversion platforms by level of intervention and approximate degree of clinical maturity.

2.6.5. Comparative table of approaches

Approach	“Reversion/re programming” mechanism	Representative examples	Development stage	Advantages	Limitations/risks
Differentiation therapy in APL [2,6,24]	Restoration of the myeloid maturation transcriptional programme via targeted degradation/inactivation of the fusion oncoprotein and reactivation of target genes	ATRA + ATO	Established clinical practice / Phase III data	High effectiveness, “functional” maturation, less myelotoxicity than chemotherapy in suitable patients	Differentiation syndrome, hepatotoxicity; disease-specific (mainly APL)
Targeted “differentiation” via metabolic/epigenetic inhibition [2,6,25]	Inhibition of mutant-IDH → reduced 2-HG → relief of hypermethylation/blockade of differentiation	Enasidenib (IDH2), ivosidenib (IDH1)	Approved / broad clinical use in AML subgroups; ongoing combination studies	Biological targeting by mutation; evidence that clinical response may be due to maturation	Differentiation syndrome (boxed warning); resistance mechanisms/clonal heterogeneity

Approach	“Reversion/re programming” mechanism	Representative examples	Development stage	Advantages	Limitations/risks
				ion and not necessarily cytotoxicity	
Epigenetic “sensitization” of differentiation [2,6,26]	LSD1 (KDM1A) inhibition reactivates the ATRA differentiation pathway in non-APL AML	TCP (tranylcypromine) + ATRA (clinical trial)	Early clinical (Phase I)	Theoretically “opens” sensitivity to differentiation in AML/ MDS subsets	Toxicities/pharmacology, possible biomarker dependence; unclear generalizability to solid tumours
Transdifferentiation of solid tumours [2,6,21]	Forcing of EMT-derived cells into post-mitotic adipocytes via manipulation of signalling/lineage pathways	MEK inhibitors + rosiglitazone (PPAR γ agonist)	Preclinical (in vivo mouse/human models)	Reduces invasion/metastasis in preclinical models; targets “plasticity” as an	Risk of incomplete conversion, patient-selection issues, pharmacotoxicities

Approach	“Reversion/reprogramming” mechanism	Representative examples	Development stage	Advantages	Limitations/risks
Chemical reprogramming to neuronal identity [2,6,22]	A cocktail of small molecules activates neuronal TFs and reduces proliferation/sphere-forming ability	U87MG glioblastoma → neuronal phenotype	Preclinical (cell lines)	Clear “state-switch” concept, possible avoidance of classic resistance	(e.g. metabolic) In vivo translation, specificity/safety, tumour heterogeneity risks
Targeted epigenetic editing (dCas9-TET1 etc.) [2,6,27]	Locus-specific demethylation/activation of tumour-suppressive networks (e.g. miR-200c) → reversal of the EMT programme	dCas9-TET1 → miR-200c (breast cancer)	Preclinical	High target specificity, theoretically fewer permanent off-target mutations	Delivery, off-target epigenetic changes, vector immunogenicity, scale-up
RNA re-wiring (miRNA mimics) [2,6,19]	Multi-target suppression of oncogenes/immune differentiation via miRNA	MRX34 (miR-34a, liposomal)	Phase I, terminated due to serious adverse	Proof-of-concept for RNA-directed	Serious immunotoxicity/SAEs, need for new

Approach	“Reversion/re programming” mechanism	Representative examples	Development stage	Advantages	Limitations/risks
	mimic		events	network remodeling	delivery vehicles/promotion schemes
Reprogramming of the TME (macrophages/STING/CD40) [2,6,28]	Shifting the balance of immune cells/inflammation/antigen presentation to “force” an anti-cancer state	CSF1R inhibitors, STING agonists, CD40 agonists	Preclinical to early clinical, variable success	Synergizes with other therapies, addresses immunosuppression	Often modest efficacy as monotherapy; systemic toxicity/targeting

2.6.6. Preclinical and clinical evidence base

The preclinical and clinical evidence base shows that the most strongly documented form of cancer reversion today is differentiation therapy in haematological malignancies. The clearest clinical example is acute promyelocytic leukaemia (APL), where the ATRA + ATO regimen essentially constitutes proof of principle that a cancer cell population can be driven to terminal differentiation and stable clinical remission, rather than being addressed only through cytotoxic elimination. [2,6,30]

Correspondingly, in AML with IDH mutations, IDH inhibitors document that “unlocking” differentiation is a feasible therapeutic mechanism, although it is accompanied by a characteristic clinical risk, mainly differentiation syndrome. [2,6,32] The same rationale extends translationally to strategies such as LSD1 inhibition to enhance the action of ATRA in non-APL AML, indicating that

reversion can arise through epigenetic “unlocking” of maturation programmes. [2,6,36]

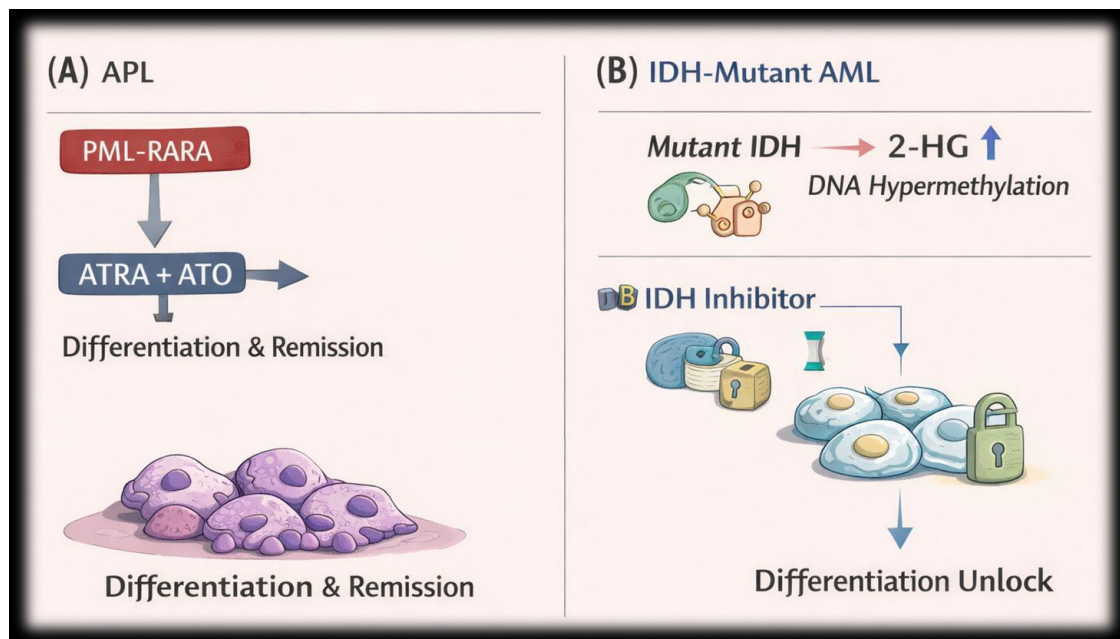


Figure 2.6.4: Clinically documented examples of therapeutic reversion through differentiation: (A) APL with ATRA+ATO and (B) IDH-mutant AML with unlocking of the differentiation block.

Beyond leukaemias, there are clinical and translational examples where the goal is not only cytotoxicity but the shifting of cellular state. Retinoids in neuroblastoma are one such example, mainly as a strategy for controlling residual disease through maturation/differentiation. In solid tumours, however, the main evidence base remains preclinical: experiments in 3D breast models show that the microenvironment and tissue architecture can impose temporary “normalization,” but this is often stimulus-dependent and does not equate to permanent reversion. [2,6,40]

Furthermore, more advanced examples such as transdifferentiation of breast cancer cells into adipocytes, targeted epigenetic reprogramming with dCas9-TET1, and RNA rewiring with miRNA tools show that state-engineering is biologically feasible, but remains limited by issues of delivery, safety, and stability of outcome. [2,6,8]

The overall picture is that therapeutic reversion is already a clinical reality in certain well-defined haematological contexts, but in most solid tumours it is still at the stage of preclinical proof of mechanism. The main conclusion is that the clearer the biological “differentiation block,” the more realistic successful reversion is; conversely, as heterogeneity, plasticity, and delivery problems increase, the strategy shifts from “true reversal” towards partial state control or chronic stabilization.

The essence is that APL is the example where the system has a clear, targetable bottleneck, so it can be moved to a new stable state. In solid tumours the system is much more multi-parametric and non-linear, so at present we mainly achieve partial remodelling rather than complete, stable reversion.

2.6.7. Translation to the clinic: biomarkers, safety, trial designs, and timelines

Translating cancer reversion/reprogramming to the clinic depends first on a strict definition of success: a transient change in morphology or markers is not sufficient; documentation is required that the cell has moved to a stable new state, one that persists after withdrawal of the intervention, is accompanied by epigenetic “consolidation,” functional restoration, and a substantial reduction in tumour-initiating capacity. [2,6,43] At the same time, biological plasticity is a double-edged sword, because partial or unwanted reprogramming can create drug-tolerant and resistant states instead of true normalization. [2,6,14]

Important clinical risks include differentiation syndrome in differentiation therapies, but also the toxicities of the delivery platforms themselves, as seen in RNA therapies such as MRX34, and this remains critical in CRISPR/epigenome editing due to delivery, off-target activity, and immunogenicity. [2,6,45][2,6,18]

Because many such interventions are cytostatic or cytodifferentiating rather than directly cytocidal, response

evaluation should not be based solely on tumour shrinkage, but on a state-monitoring framework using:

- genetic biomarkers for patient selection [2,6,47],
- transcriptional/phenotypic markers of differentiation [2,6,48],
- epigenetic readouts [2,6,49], and
- microenvironment/immunity markers [2,6,50].

At the preclinical level, demonstrating “stable reversion” requires a design with an induction phase, followed by washout/withdrawal, functional tumour-initiating-capacity assays, and single-cell multi-omics to check epigenetic consolidation [2,6,51].

At the clinical level, a realistic Phase I/II model must combine dose-finding with mechanistic confirmation (with mandatory state readouts from biopsies/marrow), and subsequently use composite endpoints, where classic markers such as ORR are supplemented by biological response, MRD, duration of response, or event-free survival. [2,6,52][2,6,56]

Finally, the most mature strategies appear to require a dual control logic: intracellular “unblocking” of the cancer programme and extracellular stabilization via the microenvironment or maintenance therapy, so as to reduce the probability of returning to the malignant attractor.

From an electrical engineer’s perspective, what matters here is not only the “control command,” but also the feedback system: we need proper biomarkers to confirm that the system has indeed moved to a new stable state and is not merely undergoing a temporary deviation.

2.6.8. Realistic timelines to the clinic, controversies, open questions, and selected references

Realistic timelines for translating cancer reversion/reprogramming to the clinic depend strongly on the type of cancer and, above all,

on whether there is reliable delivery of the intervention to the correct cellular substrate.

Based on the current level of maturity, differentiation therapies in haematological malignancies — such as APL with ATRA+ATO and IDH-targeted therapies in AML subgroups — already belong to the clinically established or immediately available field, while retinoids in selected paediatric protocols and certain forms of immune-/TME reprogramming are already in early clinical evaluation [2,6,58]. In the short-to-medium term, the most realistic development concerns optimized combinations of the epigenetic-unlock + differentiation type (e.g. LSD1i/ATRA variants), together with more precise biomarkers and possible “basket” trials in biologically plastic subsets, such as EMT-high phenotypes. [2,6,59][2,6,60]

In contrast, approaches such as precision epigenome editing, multi-gene state-engineering, and advanced RNA rewiring in solid tumours remain a medium-to-long-term goal, because they are limited by delivery, heterogeneity, and safety/off-target issues.

At the same time, the field remains open to critical scientific controversies. The most fundamental question is whether an intervention causes true, stable reversion or merely a stimulus-dependent suppression, something that directly affects the endpoints and the need for maintenance therapy. Furthermore, although there are indications that phenotype can, under certain conditions, be redirected even with genetic alterations retained, in most solid tumours polyclonality and continuous evolutionary selection favour a return to malignant attractors. [2,6,62]

Partial reprogramming is a particular risk, as it can reduce dependence on specific oncogenic pathways while enhancing resistance to therapies [2,6,63].

Ultimately, the real “rate-limiting factor” of clinical progress is delivery: as the MRX34 experience showed, even when the

biological rationale is strong, immunological interaction, pharmacokinetics, and control of off-target activity can determine the success or failure of the approach.

The critical bottleneck here is the delivery channel and stability control: it is not enough to have the correct “control signal” if we cannot deliver it selectively, at the correct intensity, without destabilizing the rest of the system.

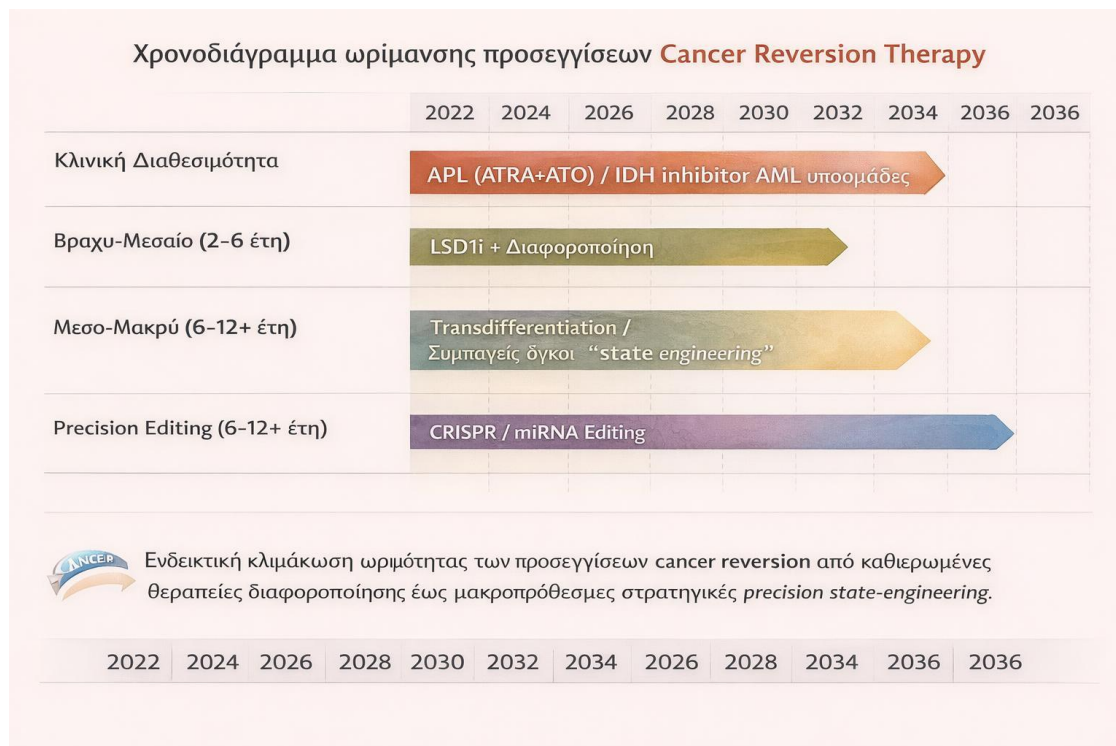


Figure 2.6.8: Indicative maturity scale of cancer-reversion approaches, from already-established differentiation therapies to long-term precision state-engineering strategies.

Chapter 3 - Technology Assessment

Chapter 3 focuses on how a “bioelectric signature” can be measured and mapped in cells and tissues, with an emphasis on non-excitable/cancer cells, where the signals are slow, low-amplitude, and often “hidden” beneath noise and interfacial artifacts. In this context, the technology assessment does not treat

the methods as “plug-and-play tools,” but as measurement chains (biological interface → transducer → front-end → filtering/DAQ → features), where reliability depends on the matching of frequency range, dynamic range, injected noise, and — above all — the sample–sensor interface. This “signal–tool mismatch” and the need for systematic noise reduction/interface optimization are explicitly described as critical obstacles to obtaining a repeatable bioelectric signature.

Based on the general principles of biomedical instrument design (measurand constraints, interfering/modifying inputs, the need for compensation and filtering), the comparison is organized around measurable criteria: sensitivity/resolution, noise, spatial resolution, sampling rate/throughput, degree of invasiveness, and compatibility with miniaturization/lab-on-chip.

Five families of technologies are compared:

- patch-clamp,
- microelectrode arrays (MEA/HD-MEA),
- impedance (EIS/impedance cytometry/EIT),
- optical voltage imaging (VSD/GEVI),
- nano/DEP/surface-charge (nanostructured interfaces, dielectrophoresis, surface-charge indicators).

The synthesis in Section 3.7 documents that, under explicit assumptions, a sensor concept aiming to read the “signature” with a view to scaling in throughput tends to favour multi-frequency impedance spectroscopy as the signal “backbone,” enhanced by DEP (pre-enrichment/separation) and/or nanostructured electrodes (improved coupling and interfacial impedance), while patch-clamp and optical voltage imaging are functionally positioned as ground truth/calibration on targeted sample subsets.

3.1. Evaluation goals/criteria

The goal of the technology assessment is to select (or hybridly compose) a technological chain — transduction → front-end → digitization → features/modeling — that maximizes discriminative capability against noise/interference and remains scalable at realistic measurement rates. This need is intensified in non-excitabile cells, where signals are small and slow, so design shifts from “spike-centric” systems to low-frequency, low-drift, ultra-low-noise readout.

Evaluation goals (engineering-oriented)

1. **Biosignal detectability:** the ability to repeatably measure a change corresponding to a biophysical parameter (e.g. V_m , C_m , R_m , ζ -potential, Cole-Cole parameters) with documented SNR and a known noise floor. (Adler, 2017; Honrado et al., 2021) [15].
2. **Spatial/temporal adequacy:** capturing either (a) a single-cell phenomenon or (b) a cohort/tissue field, with the corresponding spatial-vs-temporal trade-off (e.g. HD-MEA: sub-cellular extracellular mapping; EIT: fast but coarse tomography). (Tsai et al., 2017; Adler, 2017) [16].
3. **Biocompatibility/minimizing perturbation:** the system must not alter the measured quantity itself (e.g. VSDs can affect conductivity/function in some preparations; electrodes with high polarization can introduce drift). (Ronzhina et al., 2021; ISO 10993-1) [17].
4. **Transduction to throughput & automation:** operation of many channels/cells simultaneously, with minimal dependence on expert handling (a key requirement for a “sensor concept” that implies practical use). (Ma et al., 2023; Petchakup et al., 2017) [18].
5. **Integration into product architecture:** ability to be packaged into lab-on-chip, EMI/shielding management, compatibility with microfluidics, thermal management, and

calibration/traceability capability. (Pennati et al., 2023; Honrado et al., 2021) [19].

Prioritized criteria (quantitative & qualitative)

A practical prioritization for evaluating a bioelectric-signature sensor platform is presented below. The rationale is based on fundamental measurement/instrumentation concepts: noise budget, input impedance, bandwidth, aliasing, stability, calibration.

A. Sensitivity & SNR (highest priority) - What this means in EE terms: input-referred noise, effective noise bandwidth, 1/f corner, common-mode rejection, electrode-impedance-driven noise. - Why high priority: in non-excitable phenomena, the main limit is often SNR at low frequencies and long-term stability/distortion due to drift. (Moreddu 2023; Viswam 2019).

B. Temporal resolution / bandwidth - Determines whether we “catch” dynamic events (e.g. ms-scale spikes in MEA/optical) or slow changes (Hz-scale in impedance). (Obien et al., 2015; Villette et al., 2019).

C. Spatial resolution / localization - For mapping (HD-MEA, optical) vs. bulk signatures (EIT). (Tsai 2017).

D. Invasiveness & biocompatibility - Relates to viability, perturbation of the membrane/electrolyte, phototoxicity, electroporation, etc. (ISO 10993-1; Ronzhina 2021).

E. Throughput & scaling (automation/multiplexing) - A key criterion if the sensor concept targets screening, liquid biopsy, or large sample sets. (Ma et al., 2023; Petchakup et al., 2017).

F. Instrument cost & complexity - Not just BOM, but also space/optical requirements, shielding requirements, consumables (chips, dyes), specialized expertise. (Obien 2015).

G. Compatibility with cell types and protocols - Adherent vs. suspension, primary vs. cell lines, electrogenic vs. non-excitable, need for genetic modification (GEVIs). (Bando 2019).

H. Pathway to regulatory/safety (if a product is intended) - If the concept targets beyond research use, safety/biological-evaluation practices are required (IEC 60601-1 for medical electrical equipment, ISO 10993-1 for biological evaluation of materials). (IEC 60601-1 overview; ISO 10993-1).

3.2. Patch-clamp

Patch-clamp is the quintessential technique of intracellular electrophysiology, in which a glass micropipette forms a giga-seal (typically of $G\Omega$ order) with the membrane in order to record ion-channel currents (single-channel or whole-cell) and/or to impose a potential (voltage-clamp) or current (current-clamp). The classic methodology and the practices for achieving a giga-seal are documented in “Improved patch-clamp technique...” (Hamill et al., 1981).

In its modern form, patch-clamp extends to automated planar patch-clamp systems (microfluidic chips with apertures) that increase throughput, but introduce constraints on cell morphology/behaviour (usually easier for rounded suspension cells). (Ma et al., 2023; Gao, 2021).

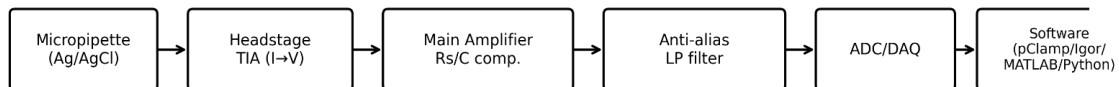
3.2.1. Performance metrics (typical ranges/characteristics)

- **Sensitivity:** single-channel currents on the order of pA; whole-cell up to tens of nA (depending on cell/channels). (Hamill et al., 1981).
- **Temporal resolution:** sub-ms achievable, limited by amplifier bandwidth/anti-alias filtering and series resistance. (Sigworth, 2006).
- **Spatial resolution:** “single-cell / single-patch” (essentially maximum locality).

- **SNR / noise:** the giga-seal reduces leakage and thermal noise; sub-pA rms noise at adequate bandwidth is reported when the seal is high-quality. (Sigworth, 2006).
- **Invasiveness:** high (mechanical contact, potential membrane disturbance, dialysis of the cytoplasm in whole-cell mode).
- **Throughput:** low in the manual form; significantly increased in automated planar patch-clamp. (Ma et al., 2023).
- **Scaling:** limited in the manual form due to human handling; better in planar form but with trade-offs. (Gao, 2021).
- **Cost:** high (low-noise amplifiers, micromanipulators, anti-vibration tables, Faraday cage, consumable pipettes).
- **Cell compatibility:** broad (excitable & non-excitable), but difficult with very small/fragile cells or with tissues without specialized techniques. (Hamill et al., 1981; Gao, 2021).

The patch-clamp front-end is essentially a transimpedance amplifier (TIA) in the headstage, with careful guarding/shielding to limit parasitics and 1/f noise. Characteristic noise measurements in integrated patch-clamp chips have been reported on the order of ~ 1 pA rms at kHz bandwidth for voltage clamp (Harrison et al., 2015) [20].

Figure 3.2-A. Patch-clamp setup: signal chain (schematic).



Typical loop: software generates Vcmd/Icmd → DAQ → amplifier; measured output returns to DAQ for acquisition.

Figure 3.2a: Typical patch-clamp chain (giga-seal interface, headstage TIA, parasitic compensation, digitization, processing).

Figure 3.2-B. Equivalent circuit of pipette-seal-membrane interface (schematic).

Elements: C_p (pipette/headstage), R_{seal} (gigaseal leakage), R_s (access), membrane modeled as $R_m \parallel C_m$.

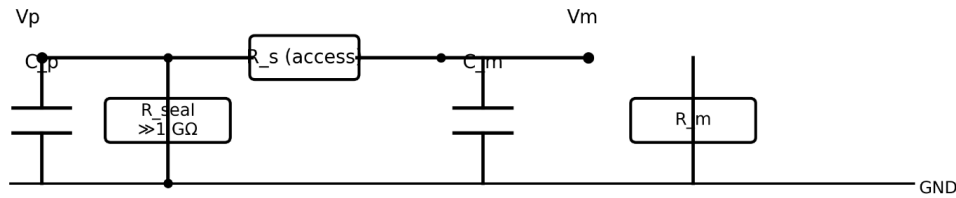


Figure 3.2b: Patch-clamp interface equivalent for noise/stability analysis (seal-limited noise and the role of access resistance).

3.2.2. Advantages

Patch-clamp is the gold standard for direct measurement of V_m /ionic currents with very high fidelity, and remains fundamental for mechanistic interpretation (channels, conductances, kinetics) and for “ground truth” in new techniques. (Hamill et al., 1981)

3.2.3. Limitations

The main drawback is low scalability and dependence on the operator (in the manual form), while the whole-cell configuration can alter the intracellular environment. Planar automation increases throughput but introduces constraints on cell preparation and does not suit every natural cell type.

3.2.4. Practical directions for hybridization with the proposed sensor concept

Patch-clamp is not treated here as a scaling technology, but as a reference method for: (i) quantifying basic electrophysiological parameters, (ii) correlating the sensor’s measured features with quantities of clear physical meaning, and (iii) establishing a benchmark for noise and front-end design. In this context, the following are proposed:

1. **Calibration and “anchoring” of features to physical quantities.** The use of classic whole-cell patch-clamp on selected cell lines and/or primary cells is recommended, so as to extract: membrane potential, membrane resistance, membrane capacitance, as well as kinetic characteristics. These quantities serve as calibration anchors for interpreting the corresponding signatures that the sensor concept will produce. [21]
2. **Parallel reference line in high-throughput scenarios.** In architectures where the primary goal is scaling (screening, multiple measurements per sample), it is proposed to examine the use of planar/automated patch-clamp as a “secondary” but standardized reference method on a small percentage of samples. This achieves control of systematic errors in the main sensor, without sacrificing the pipeline’s data-production rate. [22]
3. **Noise benchmark and transfer of readout principles to the final front-end.** Regardless of whether the final sensor is patch-based, the low-noise design principles that characterize patch-clamp systems can be used as a design template and benchmark: transimpedance amplification, guarding, minimization of parasitic capacitances and leakage paths, as well as proper handling of loop compensation/stability. The patch-clamp setup also provides a practical reference point for achieving a specific noise floor at the front-end, even if the sensor concept’s measurand is different. [23]

3.3. MEA (Microelectrode Arrays / HD-MEA)

MEAs are planar or CMOS multi-electrode implementations that mainly record extracellular potentials/currents (EAPs/LFPs) from cell populations in culture or tissues. Classically, they represent “mesoscale” electrophysiology, with the advantage of simultaneous multi-channel recording and the ability to perform long-term recordings. (Obien, 2015). [24]

HD-MEAs (high-density MEAs) exploit CMOS/microelectronics to provide tens of thousands of electrodes with small pitch, dramatically increasing spatial resolution. For example, a very large-scale CMOS MEA (65,536 channels, spacing $\sim 25.5\text{ }\mu\text{m}$, 10 kHz full-grid sampling) has been presented as a high-density extracellular recording platform. (Tsai 2017). [25]

3.3.1. Performance metrics (typical)

- **Sensitivity:** extracellular signals often on the order of tens–hundreds of μV (especially in neurons/cardiomyocytes), limited by electrode impedance and front-end noise. (Obien et al., 2015; Viswam et al., 2019). [26]
- **Temporal resolution:** ms-scale (typical sampling 10–25 kHz/channel depending on architecture). (Obien et al., 2015; Tsai et al., 2017). [27]
- **Spatial resolution:** from $\sim 200\text{--}30\text{ }\mu\text{m}$ pitch (typical/HD-MEA) up to sub-cellular EAP-propagation mapping in suitable configurations. (Tsai et al., 2017; Schröter et al., 2025). [28]
- **SNR:** strongly dependent on $Z_{\text{electrode}}$, surface modification, and front-end. The relationship between SNR and electrode impedance/geometry has been systematically analysed. (Viswam et al., 2019). [29]
- **Invasiveness:** low to moderate (extracellular, usually non-destructive), suitable for long-term recordings. (Obien et al., 2015). [30]
- **Throughput:** very high in terms of “#channels $\times$ duration,” but depends on the cell model (electrogenic vs. non-excitable). (Obien et al., 2015). [30]
- **Scaling:** high for CMOS/HD-MEA, but with system-level challenges: data rate, spike sorting, artifact rejection. (Tsai et al., 2017; Schröter et al., 2025). [31]
- **Cost:** moderate to high (MEA/HD-MEA amplifiers, chips, data handling). (Middya et al., 2021). [32]

- **Cell compatibility:** excellent for electrogenic cells (neuronal/cardiac); for non-excitable cells (e.g. many cancer cells), signals may be very low and redesign of electrodes/readout is required, as has been pointed out as a necessity for non-excitable recordings. (Moreddu et al., 2023). [15]

From an EE standpoint, MEAs are multi-channel, low-noise bio-front-ends emphasizing: input impedance, electrode offset cancellation, artifact suppression (stimulation), anti-alias filtering, and high data rates.

In the Greek academic context, analyses of MEA data (spike sorting, etc.) also appear in several theses (e.g. Evangelou, 2020). [33]

Figure 3.3-A. MEA/HD-MEA architecture and multi-channel readout chain (schematic).

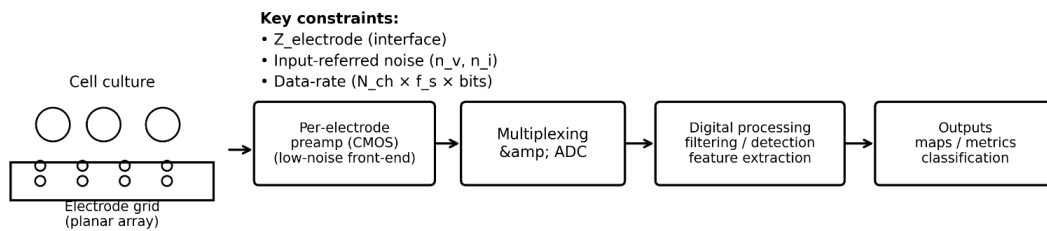


Figure 3.3-A: MEA/HD-MEA architecture and multi-channel readout chain with critical parameters: $Z_{\text{electrode}}$, input-referred noise, data rate.

Figure 3.3-B. Electrode-electrolyte interface (Randles-like) with coupling to cell source.

Elements: R_s , C_{dl} , R_{ct} , Z_W , plus coupling impedance Z_{couple} representing the cell-electrode cleft.

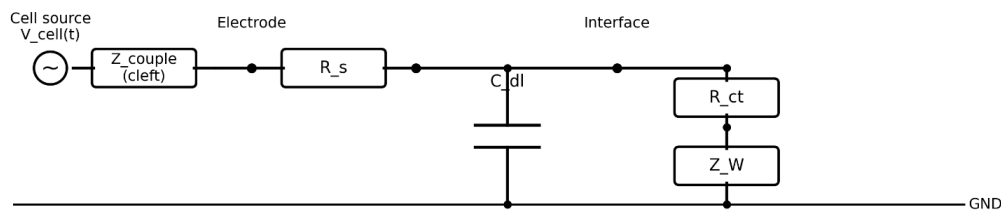
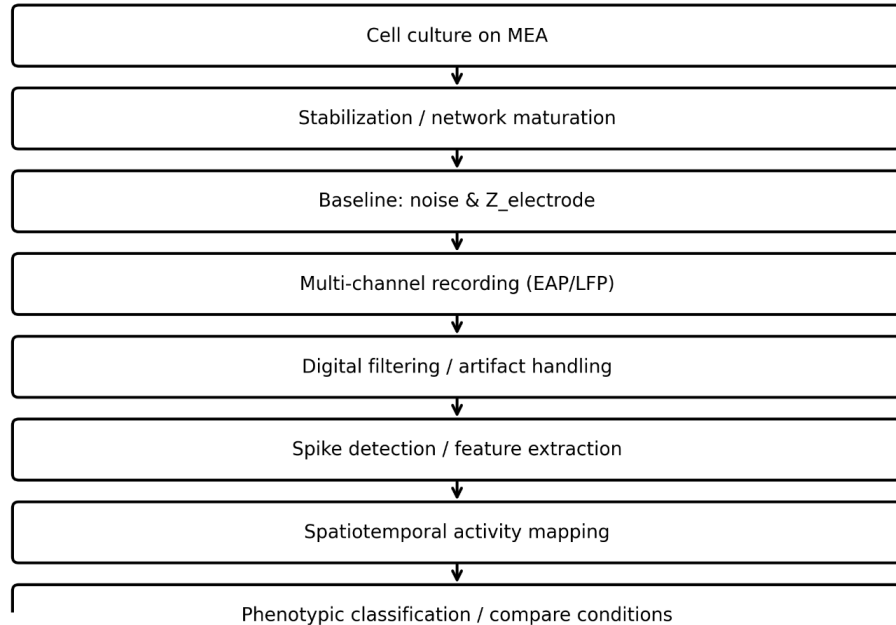


Figure 3.3-B: Electrode-electrolyte interface model (Randles-like) and its effect on the SNR of extracellular recordings.

Figure 3.3-C. MEA/HD-MEA workflow (schematic flowchart).



Flowchart 3.3: MEA/HD-MEA workflow from culture to spatiotemporal mapping and phenotypic comparison (schematic).

3.3.2. Advantages

MEA/HD-MEA systems are outstanding for parallel recording and long-term monitoring of populations, with the ability to stimulate and map. (Obien et al., 2015; Tsai et al., 2017). [27]

3.3.3. Limitations

The measurement is mainly extracellular, so recovery of absolute V_m /ionic currents is indirect; for non-excitable cells, the signal can be very small and the design must shift towards low-noise readout & interface optimization. (Moreddu 2023). [15]

3.3.4. Practical recommendations for hybridization with the proposed sensor concept

Within a hybrid methodology, MEA/HD-MEA can function as a complementary source of population-scale spatiotemporal information (cohort-level dynamics), while the sensor concept can target passive/dielectric signatures (e.g. multi-frequency impedance).

With this reasoning, the following design and experimental-integration directions emerge:

1. **Separation of roles: MEA for spatiotemporal fields — sensor concept for passive signatures.** The use of MEA/HD-MEA is proposed for capturing field patterns (e.g. synchronization, propagation, spatial organization of activity) at the population/network level, as a “contextual” measurement that complements the sensor concept’s measurements. Simultaneous or sequential data acquisition allows the correlation of spatiotemporal patterns with dielectric/impedance signatures, offering a more complete description of the phenotype under different conditions. (Obien et al., 2015) [30]
2. **Priority on reducing interfacial impedance as the main lever for SNR improvement.** The electrode–electrolyte interface (see Fig. 3.3-B) introduces impedance and noise that directly affect the quality of extracellular recordings. Especially at small electrode dimensions (HD-MEA), where impedance increases, performance improvement is achieved more effectively through reducing interfacial impedance (e.g. nanostructured surfaces, suitable coatings, increasing “effective area” without increasing geometric footprint) rather than through simple geometric scaling. This strategy is critical when the goal is stable SNR at high channel density. (Viswam et al., 2019) [29]
3. **Front-end and protocols suited to non-excitable cells: low-frequency, low drift, stable baseline.** For non-excitable cells, where changes are often slow and of small amplitude, the design of the readout chain must shift from a spike-centric profile (high bandwidth) towards low frequencies with an emphasis on stability: minimization of $1/f$ noise and drift, stable reference/grounding, robust artifact handling, and protocols that control baseline changes. This aligns with the contemporary literature that highlights an “engineering gap”

between the available tools and the requirements for detecting slow bioelectric signatures in non-excitable/cancer contexts. (Moreddu et al., 2023). [15]

3.4. Impedance / EIS / EIT (impedance spectroscopy & tomography)

The impedance “family” includes three basic categories:

1. **EIS/ECIS (Electric Cell-Substrate Impedance Sensing):** cells as an adherent layer on top of microelectrodes; the change in AC impedance correlates with adhesion, spreading, barrier function, migration, etc. (Wegener et al., 2000; Szulcek et al., 2014). [35]
2. **Impedance cytometry (microfluidic):** a single-cell dielectrophoretic/impedance spectral “snapshot” during flow, label-free, with the ability to phenotypically classify. (Cheung et al., 2005; Honrado et al., 2021). [36]
3. **EIT (Electrical Impedance Tomography):** indirect imaging of the distribution of conductivity/dielectric properties within a volume via multiple electrodes and an (ill-posed) inverse problem. (Adler, 2017; Pennati et al., 2023). [37]

3.4.1. Performance metrics

- **Sensitivity:** ECIS/EIS: high for interfacial changes (adhesion, confluence); achieves real-time monitoring of cell behaviour. (Szulcek et al., 2014). [38]
 - Impedance cytometry: single-cell differentiation/classification in a microchannel. (Cheung et al., 2005; Honrado et al., 2021). [36]
 - EIT: good for coarse tissue contrast, but sensitivity/resolution are limited by the inverse problem and electrode contact. (Adler, 2017). [39]
- **Temporal resolution:**
 - ECIS: real-time (seconds–minutes) depending on the sweep. (Wegener et al., 2000). [40]

- Impedance cytometry: high, due to flow (essentially per-cell transit time). (Honrado et al., 2021). [41]
- EIT: high (frames/sec up to tens or more, depending on hardware). (Adler, 2017; Pennati et al., 2023). [37]
- **Spatial resolution:**
 - ECIS: “effective” over the electrode area (not imaging per se). (Szulcek et al., 2014). [38]
 - Impedance cytometry: single-cell but usually as scalar/vector features (not a full image). (Honrado et al., 2021). [41]
 - EIT: lower spatial resolution compared with other imaging techniques, a known drawback. (Adler, 2017). [39]
- **SNR/stability:** strongly dependent on electrode polarization, drift, parasitics, and on the choice of 2-electrode vs. 4-electrode topology. Algorithmic processing (robust inversion) is also critical, particularly for EIT. (Brazey et al., 2022; Pennati et al., 2023). [42]
- **Invasiveness:** generally low (small AC signals), allows long-term monitoring; compatible with in-vitro monitoring. (Szulcek et al., 2014). [38]
- **Throughput & scaling:**
 - Impedance cytometry is among the most “throughput-friendly” label-free techniques. (Petchakup et al., 2017; Honrado et al., 2021). [43]
 - ECIS scales in multi-well format. (Szulcek et al., 2014). [38]
 - EIT scales with the number of electrodes/frequencies, but the inverse problem increases computational complexity. (Dimas, 2024). [44]
- **Cost:** often better than patch-clamp and optical, especially in chip-level implementations (lock-in, multiplexing). [45]
- **Cell compatibility:**

- ECIS: mainly adherent monolayers. (Szulcek et al., 2014). [38]
- Impedance cytometry: suspension/flow (suitable for blood/CTCs as a problem category). (Honrado et al., 2021). [41]
- EIT: tissues/tumours/macroscopic objects. (Adler, 2017). [39]

In practice, an EIS system consists of:

- (i) excitation (usually a current source or voltage source),
 - (ii) sense chain (differential voltage/current measurement),
 - (iii) phase-sensitive detection (lock-in or digital demodulation),
 - (iv) multi-frequency sweep,
 - (v) modelling (Cole-Cole, equivalent circuits, dielectric models).
- (Honrado et al., 2021). [41]

For microfluidic impedance cytometry, systems have been reported that extend the frequency range very high (up to hundreds of MHz in specific platforms), allowing the discrimination of sub-cellular contributions. (Haandbæk et al., 2013). [46]

Figure 3.4-A. Multi-frequency EIS/ECIS on microelectrodes with lock-in demodulation (schematic).

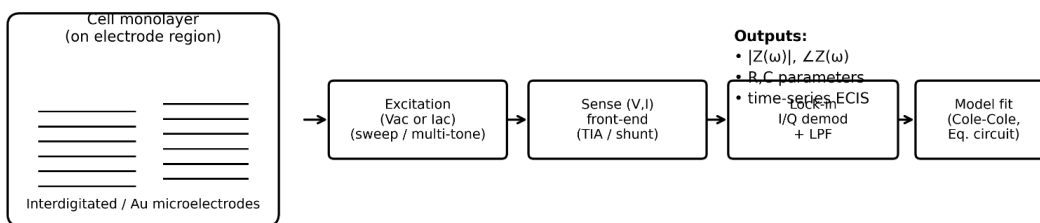


Figure 3.4-A: Multi-frequency EIS/ECIS measurement on microelectrodes (excitation, lock-in demodulation, equivalent-circuit extraction).

Figure 3.4-B. Microfluidic impedance cytometry: differential readout of single-cell transients (schematic).

Single-cell pulses at multiple frequencies → feature vector → classification/gating.

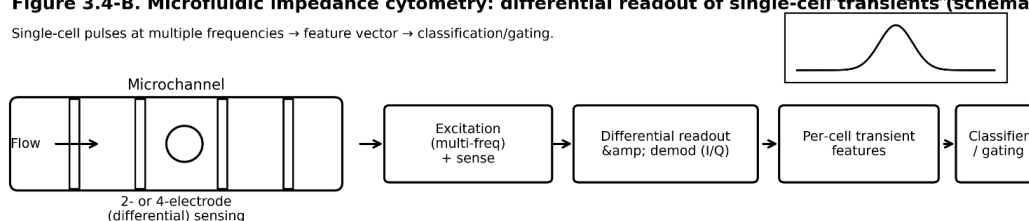


Figure 3.4-B: Impedance-cytometry configuration in microfluidics: single-cell transients → multi-frequency features → classification.

Figure 3.4-C. EIT: multi-electrode current injection, voltage measurements, and inverse reconstruction (schematic).

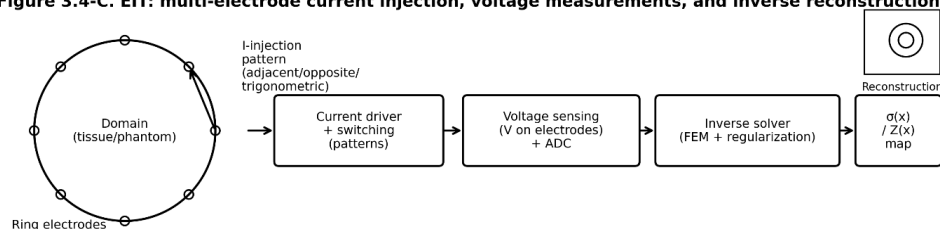
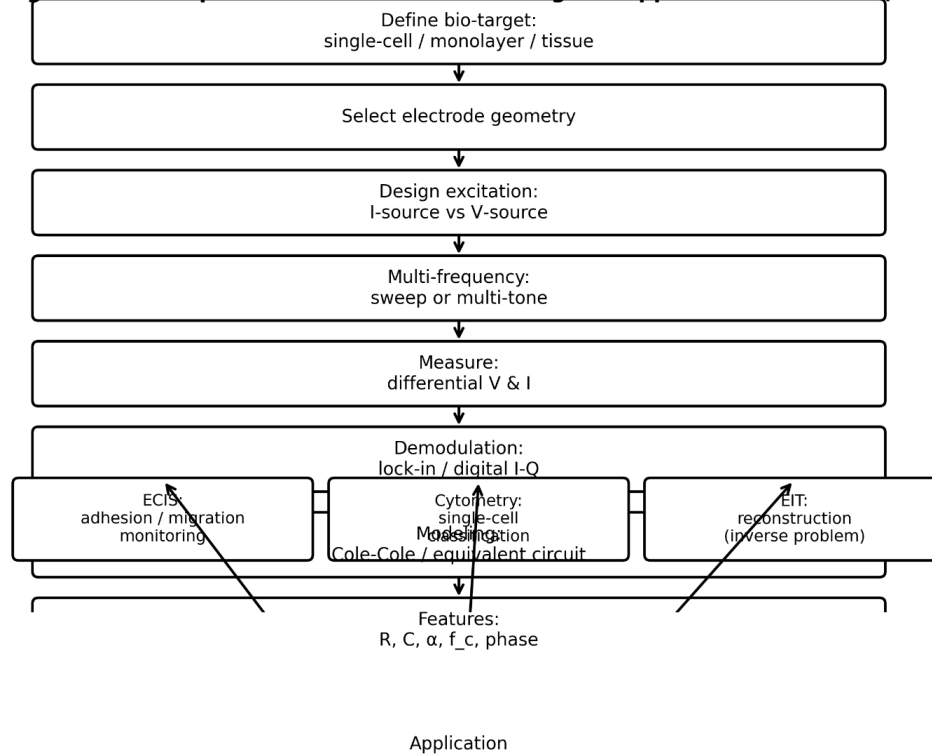


Figure 3.4-C: EIT: multi-electrode current injection and voltage measurement, solving the inverse problem for conductivity/impedance distribution.

Figure 3.4-D. Impedance measurements: design-to-application workflow (schematic).



Flowchart 3.4: Unified design/processing flow for impedance measurements: target and geometry selection → excitation (sweep or multi-tone) → differential V/I measurement → lock-in/IQ demodulation → modelling (Cole-Cole/equivalents) → feature extraction → application (ECIS, cytometry, or EIT).

3.4.2. Advantages

Impedance is among the most “EE-friendly” techniques: it converts cellular/tissue physics into a measurable transfer function, exploits mature signal techniques (lock-in, system ID), and scales naturally to chip-level implementations. In addition, it has a strong track record in real-time monitoring (ECIS) and in high-throughput single-cell phenotyping (impedance cytometry). (Szulcek et al., 2014; Honrado et al., 2021). [47]

3.4.3. Limitations

The main physical limitation is that impedance is an indirect measurement: it requires modelling/inversion in order to attribute parameters to a clear biological cause. For EIT in particular, spatial

resolution is limited and the result depends on electrode contact and regularization. (Adler, 2017; Brazey et al., 2022). [48]

3.4.4. Practical recommendations for hybridization with the sensor concept

In hybridization with a sensor concept targeting bio-/dielectric signatures, impedance spectrometry can function as the “primary” transduction mechanism, provided that the measurement chain (excitation–readout–demodulation–modelling) is designed with metrological discipline and with a clear distinction between single-cell characteristics and spatial imaging. In this context, the following are proposed:

1. **Multi-frequency parameterization instead of a one-dimensional index.** The use of multi-frequency EIS (or alternatively multi-tone excitation) is proposed so as to extract a feature vector (e.g. impedance spectrum, or Cole-Cole/equivalent-circuit parameters) instead of a scalar “R.” The multi-dimensional description reduces the risk of overfitting on narrow datasets and tends to improve generalizability across different cell lines/conditions, because it “encodes” more physical information (membrane vs. medium vs. interface contributions). (Honrado 2021) [41]
2. **Four-electrode measurement where feasible, especially in liquid/conductive samples.** For samples where electrode polarization and interfacial components (polarization impedance) are pronounced, the use of a 4-electrode (Kelvin) configuration is proposed: separating the excitation electrodes from the voltage-sensing electrodes, so as to limit the error due to polarization and improve the repeatability of the estimates. The 2-electrode version should reasonably remain an option only when packaging does not allow a four-electrode implementation. (Pennati 2023) [51]
3. **Integration of self-test and reference structures.** The provision of on-chip calibration structures (e.g. known R/C/Z

elements, “dummy” paths, reference electrodes) is proposed, so that the system supports self-test at startup, long-term drift monitoring, and verification of the correctness of the demodulation/modelling. This practice adopts a “standards-mindset” approach, in which the measurement remains verifiable/traceable even when the biological sample introduces variability that is not fully controlled. (e.g. ISO 16773-2 as an example of metrological discipline) [52]

4. **Distinct role for EIT for spatial localization — EIS/cytometry for precise characteristics.** When the goal extends to spatial mapping (cohort/tissue level), EIT can be incorporated as a secondary module for locating/distributing conductivity/impedance via inverse-problem solving. However, for “dense” and more precise single-cell characteristics (feature vectors), the main signature extraction is recommended to be based on EIS/impedance cytometry, with EIT functioning complementarily as a spatial-context tool. (Adler, 2017) [39]

3.5. Optical voltage imaging

Optical voltage imaging aims to convert V_m changes into fluorescence changes:

- **VSDs (Voltage-Sensitive Dyes):** small molecules that incorporate into the membrane; they provide very fast response, but with risks of phototoxicity/pharmacological effect. (Preuss et al., 2013; Ronzhina et al., 2021). [53]
- **GEVIs (Genetically Encoded Voltage Indicators):** protein indicators that are genetically expressed. Several classes exist (rhodopsin-based, VSD-domain based, eFRET/chemigenetic), with continuous progress in sensitivity and kinetics. (Bando et al., 2019; Kannan et al., 2022). [54]

ASAP3, for example, is reported to show a large fluorescence modulation at physiological voltages and sub-ms activation kinetics, enabling kHz-rate sampling in specialized optical methods. (Villette et al., 2019). [55]

3.5.1. Performance metrics

- **Sensitivity:** often expressed as $\Delta F/F$ per 100 mV and as single-trial detection capability. Newer tools substantially improve the signal. (Villette et al., 2019; Abdelfattah et al., 2023). [56]
- **Temporal resolution:** from sub-ms to a few ms, depending on the sensor and camera/scan. (Villette et al., 2019). [55]
- **Spatial resolution:** excellent (cellular/sub-cellular), limited mainly by optics and labeling density. (Bando et al., 2019). [57]
- **SNR:** limited by photon shot noise, scattering, photobleaching, motion artifacts, and the photophysics of the indicator itself. Recent reviews systematize noise sources/improvement strategies. (Nikolaev, 2026). [58]
- **Invasiveness:**
 - VSDs: possible pharmacological effect and phototoxicity (reports of di-4-ANEPPS effects in cardiac preparations). (Ronzhina et al., 2021; Baines et al., 2024). [59]
 - GEVIs: require genetic expression (a compatibility issue for primary/clinical samples). (Bando et al., 2019). [57]
- **Throughput:** high at the field-of-view level (many cells simultaneously), especially in in-vitro monolayers/tissues. (Bando et al., 2019). [57]
- **Scaling:** limited by optical infrastructure (microscopes, lasers, cameras, data). [60]
- **Cost:** usually high due to optics/camera/laser, especially for 2-photon/kHz. [61]

- **Cell compatibility:** VSDs more broadly, GEVIs require genetic access. (Bando et al., 2019). [57]

The chain is:

excitation source → optics (filters/dichroic) → sample (VSD/GEVI) → emission → camera (sCMOS/EMCCD) or PMT (2-photon) → motion correction → $\Delta F/F$ extraction → mapping.

Quality depends on the “photon budget,” and hence on the illumination/bleaching trade-off. (Nikolaev, 2026). [58]

Figure 3.5-A. Optical voltage imaging microscope: illumination → filters → objective → sample → emission → sCMOS → PC (schematic).

Dominant noise / artifacts:

- Photon shot noise ($SNR \propto \sqrt{N_{photons}}$)
- Photobleaching / phototoxicity
- Motion / focus drift

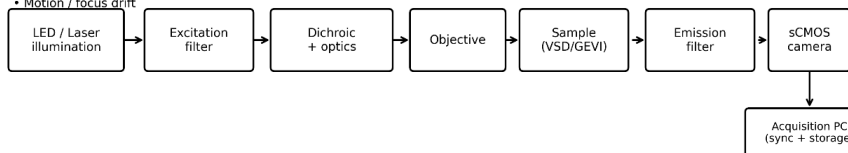


Figure 3.5-A: Typical optical voltage-imaging setup (VSD/GEVI) with critical noise points: photon shot noise, bleaching, motion.

Figure 3.5-B. Voltage imaging processing chain: raw frames → corrections → ROI traces → $\Delta F/F$ → denoise → physiological features (schematic).

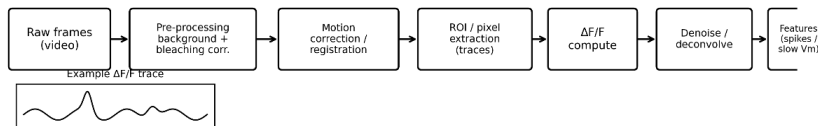
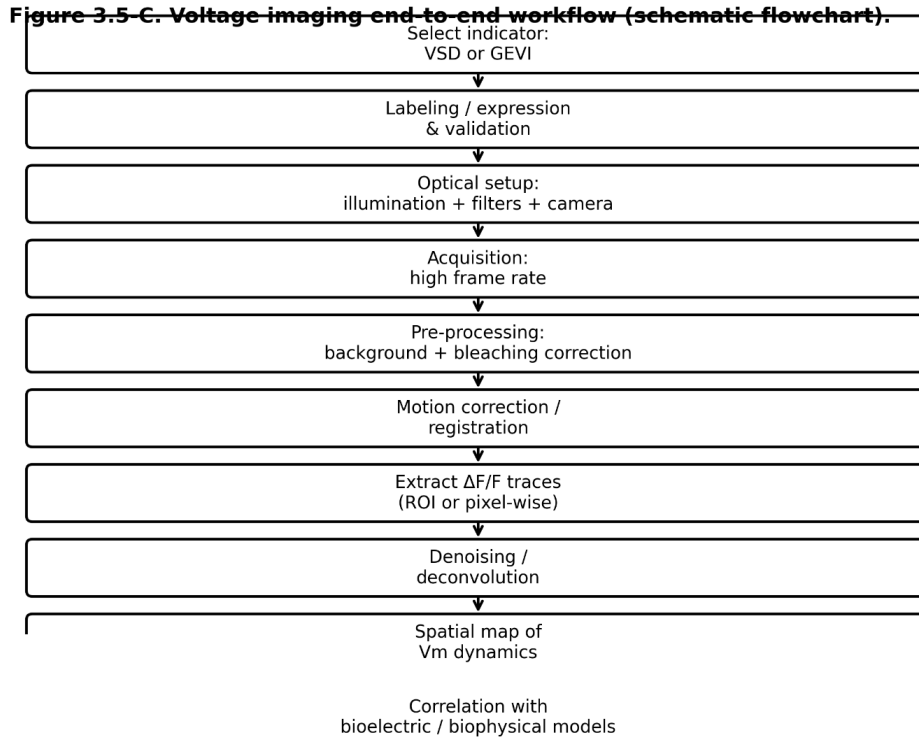


Figure 3.5-B: Signal-processing flow in voltage imaging: from raw frames to physiological features.



Flowchart 3.5: Overall voltage-imaging flow from indicator selection/expression to extraction of spatial dynamics maps and connection to electrical/biophysical models.

3.5.2. Advantages

The optical method provides parallel, high-spatial-resolution imaging of Vm dynamics, ideal for mapping and for correlation with morphology/phenotype. (Bando et al., 2019; Kannan et al., 2022). [54]

3.5.3. Limitations

The main limit is photon-limited SNR, phototoxicity, and the practical difficulty with clinical-type samples (especially for GEVIs). Also, for VSDs there is a documented need for caution because certain dyes can modify electrophysiology in some models. (Ronzhina et al., 2021). [62]

3.5.4. Practical recommendations for hybridization with the sensor concept

Optical voltage imaging (VSD/GEVI) is evaluated primarily as a spatial-reference method rather than as the main transducer of a scalable sensor. Its value, within hybrid experimental protocols, lies in its ability to produce spatiotemporal maps of changes that can be used for calibration, verification, and enrichment of the interpretation of the sensor concept's electrical features. With this functional positioning, the following directions emerge:

1. **Use as “semi-ground truth” for spatial heterogeneity and feature interpretation.** The incorporation of optical voltage imaging into targeted in-vitro tests is proposed, in order to capture spatial heterogeneity (at the ROI/pixelwise level) and create a reference for how specific dynamic patterns translate into electrical observables of the sensor concept. In this way, optical data can function as “semi-ground truth” for training/verifying the mapping between the feature space (electrical measurements) and biophysical meaning (V_m dynamics). (Villette et al., 2019) [55]
2. **Synchronized optical and electrical acquisition for time-aligned datasets and system identification.** In cases where the sensor concept is based on electrical readout (e.g. EIS/MEA), the implementation of synchronized recording (optical + electrical) with common timing and clear recording of triggers/latencies is proposed, so as to produce time-aligned datasets. This enables methodologically more robust analyses, such as system identification (correlation of inputs–outputs/response kernels) and, where warranted, causal-inference tools for understanding the relationship between optically observed changes and electrical signatures. (Bando et al., 2019) [57]
3. **Positioning VSD/GEVI as a validation module, unless the goal is purely research use.** It is recommended not to treat VSD/GEVI as the basis for “productization” without an

explicit strategy for: (i) labeling/expression (especially for GEVI), (ii) phototoxicity/bleaching, and (iii) protocol-to-protocol variation that affects reproducibility. Consequently, its practical integration is proposed mainly as a validation tool/research module, unless the sensor concept from the outset targets a research-use-only environment where these limitations are acceptable and controllable. (Nikolaev, 2026) [58]

3.6. Nano, DEP, surface charge

This section concerns “enabling technologies” that either (a) improve the interface and hence SNR, or (b) exploit electrokinetic phenomena for cell separation/characterization, or (c) target surface charge as an independent signal.

1. **Dielectrophoresis (DEP):** movement of polarizable bodies in a non-uniform electric field. Used in microfluidics for label-free separation and estimation of dielectric parameters (membrane capacitance, cytoplasm conductivity) via crossover frequencies. (Pethig, 2010; Wang et al., 2000). [65]
2. **Surface charge / ζ -potential:** the net surface charge affects interactions and can differ between cancer and normal cells; approaches exist that quantitatively measure surface charge with electrochemical devices. (Hu et al., 2025; Baca et al., 2022). [66]
3. **Nanoelectrode arrays & nanoscale FETs:** nanostructured electrodes/nanotubes/nano-FET probes for stronger coupling, up to intracellular-like recordings from arrays (thousands of sites). (Abbott et al., 2019; Xie et al., 2012; Duan et al., 2011). [67]
4. **Nanotechnology as a “bridge” of bioelectricity-translational tools:** a summary literature position that nanosystems can increase throughput/precision and make “bioelectric signatures” measurable in non-excitable cells. (Moreddu 2023). [15]

3.6.1. Performance metrics (typical)

- **Sensitivity:**
 - DEP: depends on $\text{Re}\{K(\omega)\}$ and $\nabla|E|^2$; capable of discriminating subpopulations and cell-line stages in studies. (Liang et al., 2020). [68]
 - Surface charge: quantitative measurement of surface charge with electrochemical setups has been proposed as non-invasive in a cyto-compatible environment. (Hu et al., 2025). [69]
 - Nanoelectrodes: the ability to perform intracellular recordings at a very large scale (thousands of neurons) has been reported with nanoelectrode arrays. (Abbott et al., 2019). [70]
- **Temporal resolution:** DEP/crossover and surface-charge assays are usually seconds–minutes; nanoelectrodes can reach ms/Hz–kHz depending on readout. (Abbott et al., 2019; Pethig, 2010). [71]
- **Spatial resolution:** DEP/ ζ -potential is typically “per cell” or population-level; nanoelectrode arrays can achieve high site density. (Abbott et al., 2019). [70]
- **SNR & repeatability:** determined by electrode stability, (Joule) heating, electroosmotic flows, and by control of medium conductivity (DEP is sensitive to medium composition). (Yao et al., 2024). [72]
- **Invasiveness:**
 - DEP: usually low if fields/frequencies are within safe limits, but requires viability control. In DEP separation for cancer cells, the need to preserve viability has been noted. (Varmazyari et al., 2022). [73]
 - Nanoelectrodes: may require electroporation/penetration (hence moderate invasiveness). (Xie et al., 2012). [74]
- **Throughput:** DEP microfluidics can achieve high throughput and is suitable for label-free separation/early

detection as a family of solutions. (Najafipour et al., 2025). [75]

- **Scaling:** good for microfabrication, but requires careful design of electrodes/3D fields and power. (Yao et al., 2024). [72]
- **Cost:** moderate (microfabrication/driver electronics) relative to optical, but can increase for advanced nanoarrays. [76]
- **Cell compatibility:** high in theory (label-free), but DEP often requires special conductivity buffers; ζ -potential and electrochemical measurements require careful chemistry/ions. (Pethig, 2010; ASTM E2865). [77]

For DEP one needs: an AC source/driver (often 10 kHz–MHz), microelectrode geometry, microfluidic control, and imaging or electrical readout to estimate trajectories/crossover. (Pethig, 2010; Yao et al., 2024). [78]

For ζ -potential, there are established measurement practices and guides for biological nanomaterials that are carried over as a philosophy (e.g. ASTM E2865). [79]

For nanoelectrode arrays/nano-FET probes, the critical factor is the nano-interface and the ability to multiplex, aiming at high-density “intracellular access.” (Abbott et al., 2019; Duan et al., 2011). [80]

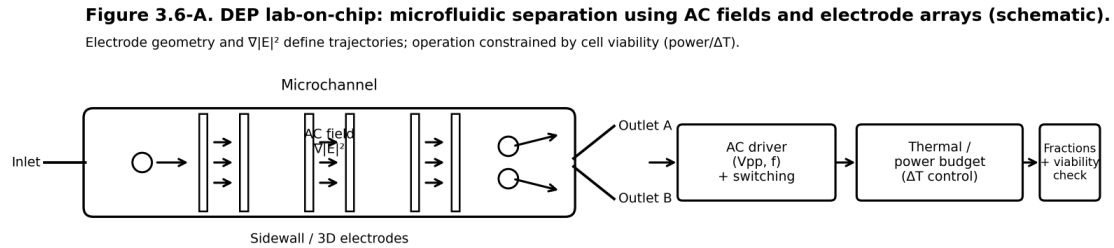


Figure 3.6-A: DEP-based microfluidic separation: electrodes, field, flow lines, fraction collection under viability constraints.

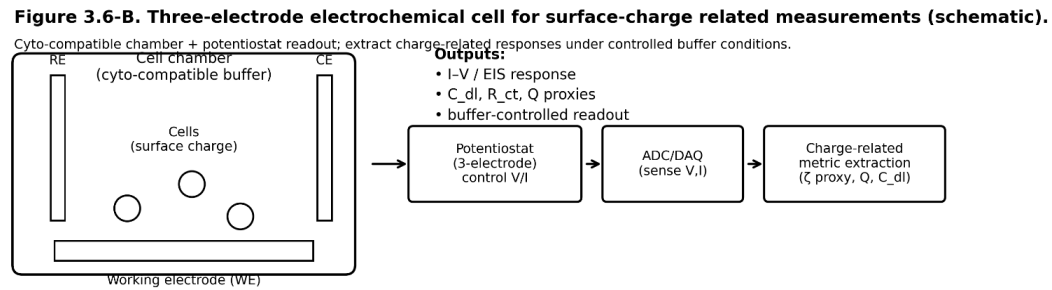


Figure 3.6-B: Three-electrode electrochemical setup for quantitative measurement of cell surface charge in a cyto-compatible environment.

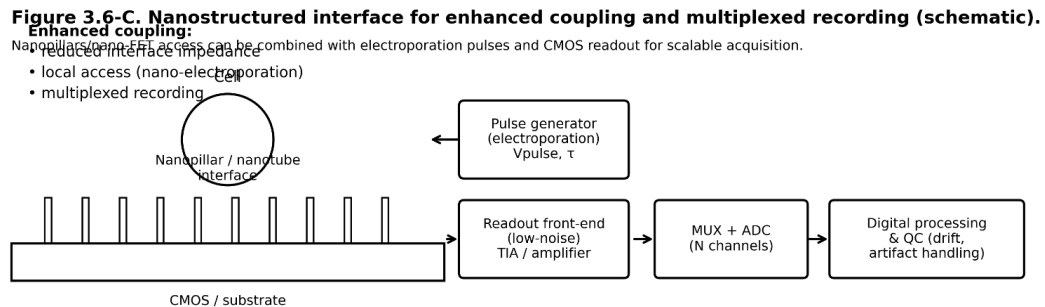
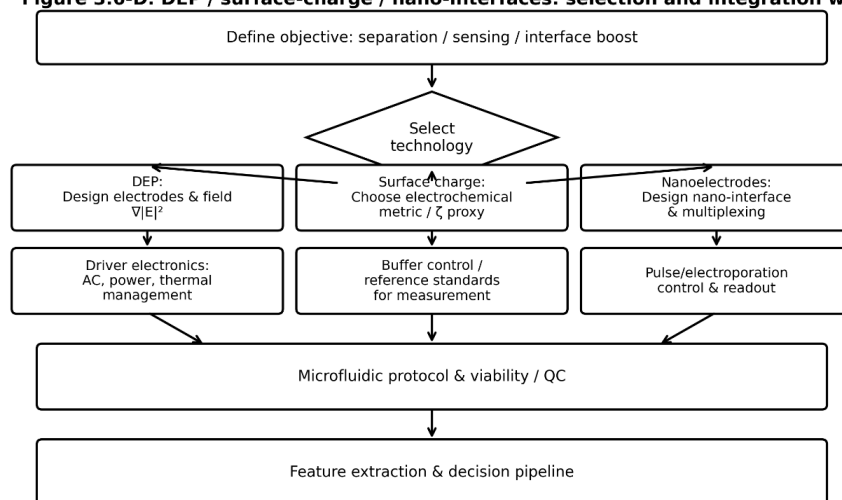


Figure 3.6-C: Nanostructured interface for enhanced coupling (e.g. nanopillar electroporation / nano-FET access) and multiplexed recording.

Figure 3.6-D. DEP / surface-charge / nano-interfaces: selection and integration workflow (schematic).



Flowchart 3.6: Selection and integration flow for DEP / surface-charge / nano-interface technologies: from goal definition (separation/sensing/interface boost) to electrode/field or metric design, to electronic driving/control, to the microfluidic protocol & viability/QC, and finally to feature extraction and the decision pipeline.

3.6.2. Advantages

DEP and nano-interfaces are particularly attractive because: (i) they offer label-free handling/enrichment, (ii) they can be integrated into lab-on-chip, (iii) they naturally provide “electrical” characteristics (crossover → membrane capacitance), and (iv) nanoelectrode arrays show that scaling intracellular readout is feasible at a scale not achievable with classic patch-clamp. (Abbott et al., 2019; Najafipour et al., 2025). [76]

3.6.3. Limitations

DEP is sensitive to medium conductivity/thermal effects, and repeatability depends on strict standardization. Nano-interfacing often requires specialized fabrication and can increase complexity/cost. ζ -potential/surface charge has biological/chemical dependencies (pH, ionic strength), and therefore requires standardized metrological discipline. (ASTM E2865; Yao et al., 2024). [81]

3.6.4. Practical recommendations for hybridization with the sensor concept

DEP technologies, nanostructured interfaces, and surface-charge measurements are not treated as “competitors” to the main transducer, but as enhancing subsystems that can improve (i) the quality

of the sample at the input, (ii) the observability of electrical parameters, and (iii) SNR via interfacial engineering. Within a hybrid architecture, the following are proposed:

1. **DEP as front-end sample conditioning (separation/enrichment before the main sensor).** The use of DEP as a sample pre-processing stage is proposed, so that the main sensor is fed with a cleaner and/or enriched fraction of the original population. This approach is particularly compatible with blood scenarios or mixed cell populations, where impurity/heterogeneity increases the variance of electrical signatures and burdens classification. The functional role of DEP here is to transfer part of the “difficulty” from the downstream feature space to an upstream physical-separation stage. [85]
2. **Coupling DEP-spectroscopy with EIS to reduce modelling uncertainty.** The combined use of DEP-spectroscopy with multi-frequency EIS is proposed, so as to provide an independent estimate of dielectric/membrane properties. Methodologically, this limits uncertainty and non-uniqueness in fitting EIS equivalent circuits, because part of the parameters is “anchored” by the DEP response. [86]
3. **Nanostructured electrodes as an “SNR booster” via reduction of interfacial impedance and increased coupling.** In cases of very small signals, the investigation of nanostructured electrodes/nanostructured interfaces is proposed as a way of effectively reducing the interface impedance and enhancing cell–electrode coupling. This

design aims to improve input-referred SNR without necessarily increasing geometric footprint. [70]

4. **Addition of a surface-charge channel as a complementary feature-space dimension, with strict standardization of conditions.** If additional “orthogonal” information relative to impedance is sought, the incorporation of measurements related to surface charge / ζ -potential as a separate feature channel is proposed. For the measurements to be comparable and reproducible, strict control of medium (buffer) conditions is required, especially pH, ionic strength, and conductivity, as these parameters directly affect the double layer and electrochemical responses. [87]

3.7. Comparative table & conclusion for the sensor concept

No constraints/design were given, so the following assumptions are adopted:

- 1) The sensor targets a bioelectric signature (a state vector of parameters, not a single measurement).
- 2) It needs practical scalability (at least medium throughput), with a view to lab-on-chip. [88]
- 3) It requires low invasiveness and applicability to non-excitable cells, where the signal is small and the main difficulty is noise/stability. [15]
- 4) It is not assumed that optical/genetic access already exists, so techniques requiring genetically encoded reporters are considered optional.

3.7.1. Scoring scale and criteria weights

A 1–5 scale is used: - 1 = very weak/not suitable, - 3 = moderate/depends, - 5 = excellent.

The weights are assigned with EE reasoning (emphasis on SNR, invasiveness, throughput) and correspond to:

- Sensitivity/SNR: 20%,
- Temporal resolution: 10%,
- Spatial resolution: 10%,
- Invasiveness: 15%,
- Throughput: 15%,
- Scaling: 10%,
- Cost: 10%,
- Compatibility: 10% (to connect with the previously mentioned “engineering gap,” as well as the need for low noise in non-excitables signals).

3.7.2. Comparative scoring table

Table 3.7-A: Comparative scoring of techniques against EE criteria. The scores are briefly justified in the text (e.g. patch-clamp: excellent SNR but low throughput; EIS/cytometry: excellent throughput and good compatibility with lab-on-chip). [89]

Μέθοδος	Ευαισθησία/SNR	Χρονική	Χωρική	Επεμβατικότητα	Throughput	Κλιμάκωση	Κόστος	Συμβατότητα	Σταθμισμένο σκορ
Patch-clamp	5	5	5	1	1	1	2	3	2.9
MEA/HD-MEA	3	4	4	4	4	4	3	3	3.6
Impedance (EIS/ECIS/Impedance cytometry/EIT)	4	3	3	5	5	4	4	4	4.1
Optical voltage imaging	4	4	5	3	4	3	2	3	3.55
Nano/DEP/surface-charge	4	3	4	3	5	4	3	4	3.8

Table 3.7-A: Comparative scoring of techniques against EE criteria.

3.7.3. Justification of the critical scores

- Patch-clamp scores maximally on fidelity (sensitivity/temporal/spatial) due to its gold-standard character, but low on throughput/scaling due to operator dependence; this is also documented by the field’s shift towards automated patch-clamp to increase throughput. (Hamill et al., 1981; Ma et al., 2023). [90]

- MEA/HD-MEA has high scaling/multi-channel capability and good non-invasiveness, but in non-excitable applications SNR can be limiting without specialized interface/readout optimization. (Obien et al., 2015; Moreddu et al., 2023). [91]
- Impedance (EIS/ECIS/impedance cytometry) excels in throughput and compatibility with a product form factor (lab-on-chip), while EIT adds imaging capability with lower spatial resolution; overall it offers the best “engineering-to-product” balance. (Honrado et al., 2021; Adler, 2017). [92]
- Optical voltage is top-tier spatially and very strong temporally, but has high cost/complexity and labeling/phototoxicity limitations. (Bando et al., 2019; Ronzhina et al., 2021). [93]
- Nano/DEP/surface charge scores very well on throughput/scaling and as an “interface booster,” but introduces new complexities (drivers, field uniformity, buffer control). (Varmazyari et al., 2022; Yao et al., 2024). [94]

3.7.4. Proposed comparative table of metrics

Table 3.7-B: Summary mapping of “what each technique measures” and what its physical scale/rate is. The ranges are indicative and depend on protocol/instrumentation. [95]

Τεχνική	Πρωτεύον observable	Κλίμακα	Τυπική χρονική κλίμακα	Κύριο bottleneck
Patch-clamp	$I(t)$, $V_m(t)$, channel kinetics	single-cell/patch	sub-ms έως s	χειριστής, invasiveness
MEA/HD-MEA	extracellular $V(t)$	network/culture	ms (kHz sampling)	coupling & $Z_{\text{electrode}}$, non-excitable SNR
EIS/ECIS	$Z(\omega)$, ισοδύναμα R/C/a	electrode-region/monolayer	s–min (sweep)	μοντελοποίηση & drift
Impedance cytometry	single-cell impedance features	single-cell in flow	ms (transit)	buffer control, feature mapping
EIT	χάρτης αγωγιμότητας/εμπέδησης	tissue/volume	frames/sec	ill-posed inverse, spatial resolution
Optical voltage imaging	$\Delta F/F \leftrightarrow V_m$	cell/subcellular	ms–sub-ms	photon-limited SNR, phototoxicity
Nanoelectrode arrays	electrical intracellular-like signals	multi-cell array	ms–kHz	fabrication, interface reliability
DEP/surface charge	trajectory/crossover, ζ -potential	single-cell/population	s–min	medium dependence, standardization

Table 3.7-B: Summary mapping of what each technique measures.

3.7.5. Final conclusion and recommendation for the sensor concept

Based on the above assumptions and weights:

The most balanced direction for an unspecified “sensor concept” that seeks to reveal a bioelectric signature with scalability is an architecture with multi-frequency impedance spectrometry (EIS/impedance cytometry) at its core, because:

- it is inherently label-free and suitable for non-excitable phenotyping,
- it scales well in microfabrication (many sites or high flow rate),
- it exploits mature EE techniques (lock-in, system identification, equivalent circuits),
- it “fits” naturally into lab-on-chip/point-of-need devices, as is also highlighted by recent reviews of impedance cytometry and ECIS. (Honrado et al., 2021; Moghtaderi et al., 2024).

[96]

3.7.6. Brief proposal for a “system architecture” (independent of constraints)

- **Front-end:** differential low-noise instrumentation, lock-in (analogue or digital IQ), multi-tone excitation.
- **Transducer:** microelectrodes (interdigitated or 4-terminal), with nano-coating for Z reduction.
- **Sample handling:** DEP-based enrichment or microfluidic focusing → impedance cytometry + EIS station.
- **Model layer:** equivalent circuit + dielectric model + ML classifier (with patch-clamp/optical labels on a training subset). (Honrado et al., 2021; Abbott et al., 2019). [99]

The above recommendation is “best-effort” because the sensor concept is unspecified; if the concept ultimately has a very specific biological target (e.g. V_m only, or adhesion only), then the weights/scores should be readjusted accordingly.

Chapter 4 - Product Design

Passive biopotential “V_m-like” + Active impedance (EIS/ECIS/impedance cytometry)

The sought-after “hybrid biosignal meter” combines two fundamentally different measurement chains on the same chip/electrode arrangement:

- (a) passive recording of extremely low-amplitude, low-frequency signals (V_m-like, slow field/biopotential changes), and
- (b) active measurement of complex impedance (EIS/ECIS/impedance cytometry) via excitation (AC voltage or AC current) and synchronous demodulation (lock-in/IQ).

The main challenge is that passive recording requires ultra-low-noise + low-drift behaviour near DC, while active impedance

spectrometry requires clean excitation, high linearity, control of currents/fields (cell viability), and often MHz-range response.

The need for “hybridization” is also described in the uploaded report as a central engineering gap for non-excitabile cells (signals much smaller than action potentials), and a backbone architecture is proposed: a differential low-noise front-end + lock-in/IQ + multi-tone excitation, with microelectrode transducers (interdigitated or 4-terminal), possible integration of microfluidics/DEP, and a final modelling layer of equivalent circuits/dielectric models and classification.

At the specification level, the following are proposed (indicatively/as ranges, where not otherwise fixed): for V_m-like mode, an amplitude target on the order of 1–100 μ V (alternative scale of 10–500 μ V if extracellular-like signals are desired), bandwidth 0.01–300 Hz, and an SNR target that must be explicitly stated (target bands proposed from ≥ 6 dB at the minimum signal up to ≥ 20 –40 dB).

For impedance mode, the target is a 1 kHz–10 MHz sweep (the minimum for cytometry/dielectric information often falls in the hundreds of kHz–tens of MHz range), with excitation typically from 10–100 mV (ECIS/tissue cultures) up to 0.1–0.5 V, and documented examples of even higher amplitudes in specific configurations — however, with a “patient-like” mindset, a conservative ceiling and explicit control of the E-field are recommended. [1]

The chapter concludes with: (i) a proposed electronic-architecture block diagram (Fig. 4.1) and a baseline hybrid block diagram, (ii) electrode comparisons/selection with an indicative IDE+reference+guard layout (Fig. 4.2), (iii) a mode-control flowchart for interference control, and (iv) Table II-1 of “final specs” with explicit marking of the TBD parameters in Table II-2 (N_{ch}, SNR targets, assay type: adherent vs. suspension, etc.)

4.1. Problem framework and requirements for the hybrid meter

As mentioned above, in non-excitable (cancer) cells the interest shifts from “spike-centric” recordings towards slow/static bioelectric observables, with requirements for low input noise and long-term stability; indeed, orders of magnitude of activity at the cohort level of $\sim 0.1\text{--}100$ pA and $1\text{--}10$ Hz are noted, so the architecture must lower the noise floor and drift into ranges that many commercial MEA tools do not target without redesign.

At the same time, impedance-based spectroscopy (EIS/ECIS and especially impedance cytometry) is a strong “backbone” because it gives access to membrane polarization, the double layer, cytoplasmic conductivity, and (at higher frequencies) information on sub-cellular structures, in high-throughput form. [2]

From a product standpoint, the hybrid sensor must be defined as a system of two temporally/functionally distinct modes that share part of the interface (electrodes, microfluidics, common ground/reference, EMI shielding), but have different analogue front-ends:

- **Passive V_m -like mode:** ultra-high input impedance, very low input-referred noise at $0.01\text{--}300$ Hz, a mechanism for handling DC offsets/electrochemical drift, strong CMRR, and bias return without “loading” the signal.
- **Active impedance mode:** an excitation path (V-source or I-source), a current/voltage sense path with linearity and calibration, lock-in/IQ demodulation, and (for cytometry) bandwidth/sampling sufficient to capture cell-transit pulses at high throughput. [3]

4.2. Target signal specifications and functional modes

4.2.1. Passive biopotential Vm-like mode

- **What we practically measure:** without patch-clamp, the “Vm-like” signal on chip electrodes tends to be an extracellular potential/field or slow changes in electrochemical potential related to ionic flows/adhesion/collective fields, rather than a direct intracellular Vm. The measurement target here is the external electrical trace.
- **Signal amplitude target (as a requirement):**
 - Proposed baseline range: 1–100 μV_{rms} , so as to cover “very small” signals.
 - Alternative, more “MEA-like” range for extracellular recordings: 10–500 μV (classic extracellular recording reports typically cite tens–hundreds of μV). [4]
- **Bandwidth:**
 - Proposed for slow-Vm/slow changes: 0.01–300 Hz.
 - Note: in many ECG/EEG-type biopotential front-ends, typical bandwidth is 0.05–100 Hz for the “clinical-type” example, which is compatible as a reference point for filter/CMRR targets. [5]
- **SNR targets:**
 - For “detecting changes” at the 1–10 μV level, a realistic target is $\text{SNR} \geq 6\text{--}10\text{ dB}$ at the minimum signal and $\geq 20\text{ dB}$ at the typical signal (implying input-referred noise $\sim 1/10$ of the typical signal in the corresponding ENBW).
 - For “quantification” (parameter estimation) of longitudinal changes, a target of $\geq 20\text{--}40\text{ dB}$ in the band of interest is recommended.

4.2.2. Active impedance mode (EIS/ECIS/impedance cytometry)

- **Frequency target (frequency plan):**
 - Baseline requirement: a 1 kHz–10 MHz sweep.
 - For impedance cytometry, different sub-cellular features “appear” in different bands: below ~100 kHz the double layer dominates, 0.1–1 MHz relates to electrical size/membrane shielding, 2–10 MHz to membrane polarization/morphology, 10–30 MHz to internal conductivity/organelles (depending on cell size and medium conductivity). [6]
 - For ECIS/monolayers, it is common to work from lower Hz up to the kHz range (depending on model/target), but on microelectrodes the Helmholtz double layer and polarization raise the low-frequency impedance and reduce SNR — for this reason, careful selection of low frequency and/or use of a 4-electrode scheme is often applied. [7]
- **Excitation amplitude target & cell viability (safety requirement):**
 - For tissues/models (e.g. 3D cultures), applied voltages of 10–100 mV are reported as “applicable” orders of magnitude, depending on electrodes/medium/cells. [8]
 - In microfluidic settings, due to significant voltage drop across the microchannels, the potential difference can reach 0.1–0.5 V. [8]
 - In the impedance-cytometry literature, examples appear with amplitudes from 0.1 V to 0.5 V, but also cases up to 5 V (depending on geometry/target/settings), while frequencies such as 500 kHz and 10 MHz are common for multi-parametric information. [9]

Practical conclusion for design:

- Define a conservative default excitation (e.g. 10–100 mV for adherent ECIS, 0.1–0.5 V for cytometry) and incorporate a programmable ceiling with interlocks.
- Translate the excitation into an electric field ($E \approx V/d$) and set an E limit so as to avoid unwanted changes in V_m /channels: e.g. it is reported that fields of 1–10 V/cm can change V_m by ~ 1.5 –15 mV in a 10 μm -radius cell, so it must be ensured that the applied field does not exceed the desired protocol limits. [8]
- To avoid electroporation, in pulsed applications fields from 0.1 kV/cm and above are reported as the region of interest per strength–duration, so EIS fields must remain much lower than such scales (unless the goal is deliberate perforation). [10]

Impedance SNR/accuracy:

- For product-level metrology, convert SNR into $|Z|$ and $\phi(Z)$ error (e.g. $\leq 0.5\%$ magnitude, $\leq 0.5^\circ$ phase as an initial engineering target) and into repeatability per well/channel. The requirement will depend on the feature extraction (Cole-Cole/equivalent-circuit fitting).

4.3. Electrode interface and sensor geometries

The choice of electrodes is “half the instrument”: it determines electrode impedance, polarization, 1/f noise, spatial selectivity, positional dependence (in flow), and the ability to perform a 4-terminal measurement to reduce interfacial errors. [11]

4.3.1. Comparison of basic electrode configurations (Table II-1)

Configur- ation	Typical use- case	Advantages	Disadvantage s/risks	Conseq uences for Nch
IDE (Interdi	ECIS/adhesion/ proliferation on	Large effective	Output is a “sum” over	Often 1–16

Configuration	Typical use-case	Advantages	Disadvantages/risks	Consequences for Nch
Agitated Electrodes)	adherent monolayers	perimeter → increased sensitivity to surface changes; simple 2-terminal implementation; suitable for kinetic monitoring of adhesion/morphology. [12]	many cells (requires density/place ment) and difficulty with low-density signals. [13] Also strong influence of the electrode–electrolyte interface at low frequencies. [8]	“zones” /chip for wells/patches; scalable to arrays, but not the same philosophy as MEA.
Microchannel 2-electrode / 4-electrode (facing or coplanar)	Impedance cytometry (suspension), μ -EIS on trapped cells	For cytometry: multi-frequency fingerprints in the 0.1–30 MHz range with high throughput. [6] With 4-electrode configuration, the	Position dependence in coplanar geometries (cell height) and need for focusing/compensation. [14] At very low f, the risk of Faradaic/double-layer effects	Usually 1–8 sensing zones per microchannel; multizone capability → Nch increases

Configuration	Typical use-case	Advantages	Disadvantages/risks	Consequences for Nch
		influence of the electrode/medium on the output is reduced. [7]	increases. [15]	quickly (e.g. 8–64).
MEA grid (multi-electrode array)	Spatial mapping of extracellular signals and/or local impedance	Parallel recording of many locations; typical extracellular signals are tens–hundreds of μV in excitable cells, so it can function as a platform/reference for low-noise readout. [16] Scaling to large arrays (thousands) in specialized CMOS-MEA	For non-excitable/slow signals, the bottleneck is drift and 1/f noise; also electrode impedance/cell coupling is decisive. [18] High complexity of routing, MUX, data bandwidth.	Nch from 1 up to 128 (or much higher in HD-MEA). For a general-purpose product, 16–64 is a practical initial scale; the report does not set a limit.

Configuration	Typical use-case	Advantages	Disadvantages/risks	Consequences for Nch
		contexts. [17]		

4.3.2. Proposed strategy for selecting Nch

Precise number of channels (Nch):

- **Adherent ECIS/IDE:** Nch is linked to the number of wells/patches (e.g. 1–16).
- **Microfluidic cytometry:** Nch = number of zones × (channels per frequency/IQ). In practice, 4–32 very quickly. [6]
- **MEA-like mapping:** Nch scales as a grid (1–128 with no hard limit).

Interdigitated (or 4-terminal microelectrodes) + DEP-based enrichment/microfluidic focusing is consistent with an architecture that starts with $N_{ch} \approx 8\text{--}32$ for a first implementation and anticipates expansion to 64–128 with repeating tiles (repeated AFE+MUX blocks).

4.4. Analogue Front-End and impedance excitation

4.4.1. AFE requirements for the passive Vm-like mode

Input impedance: - Target: $\geq 1\text{ G}\Omega$ differential (or equivalent very low input conductance), so as not to load the electrode impedance and to limit bias-current-induced offsets. A practical reference is that a classic multichannel biopotential AFE such as the ADS1299 states a DC input impedance of $\sim 1000\text{ M}\Omega$ (without lead-off) and input bias currents in the pA range. [19]

CMRR: - Target: $\geq 100\text{ dB}$ at 50/60 Hz as a baseline; high-quality AFEs show CMRR of $\sim 110\text{--}120\text{ dB}$ at 50/60 Hz. [20] - The

“patient-like” philosophy imposes control of common-mode coupling and proper biasing/shielding.

Noise & drift/offset tolerance: - For 0.01–300 Hz, the critical parameter is input-referred noise and the 1/f corner. Example: the ADS1299 gives input-referred noise (0.01–70 Hz) of approximately 1 μV_{pp} (at specific gain/data-rate settings). [19] - Offset drift target (indicative): $<0.1\text{--}1 \mu\text{V}/^{\circ}\text{C}$ at the front-end (or equivalent with a servo-loop), so that slow changes are not “buried.” Indicatively, the ADS1299 reports an offset drift of $\sim 80 \text{ nV}/^{\circ}\text{C}$. [19] - Not specified: the allowable drift in $\mu\text{V}/\text{hour}$ for the protocol (cell assays may run for hours–days).

Filters (HP/LP) and DC offsets: - For electrochemical/electrode offsets, it is classic practice to use AC coupling and/or “DC suppression” so as not to saturate the AFE. In ECG analogue front-ends, a design target is reported of rejecting offsets up to $\pm 300 \text{ mV}$ while simultaneously amplifying the small signal. [21] - Proposed filter: HP 0.01–0.05 Hz (configurable) and LP 300–500 Hz, analogue and/or digital, with a 50 Hz notch preferably implemented digitally (so as not to introduce analogue ringing on slow signals). [22]

Input protection & ESD: - Even in an in-vitro setup, the input must withstand ESD/handling. As a practice, we design for IEC 61000-4-2 up to Level 4 ($\pm 8 \text{ kV}$ contact, $\pm 15 \text{ kV}$ air). [23] - Protection includes: series resistors (without excessively increasing Johnson noise in the band), clamp diodes/TVS arrays, and a large-value “bias return” so that the input does not remain floating (especially when MUX is used). The report notes this explicitly as a practical need (“bias return”).

4.4.2. Excitation for EIS: I-source vs. V-source, sweep strategies, waveforms

Current source vs. Voltage source (basic choice):

- **I-source (constant current):** Typical in ECIS/TEER approaches. In an ECIS review, the use of a constant AC current of $\sim 1 \mu\text{A}$ is reported, with impedance computed from mV-level voltage changes. [24] It is advantageous when we want stable excitation independent of Z , but requires a current source with high output impedance across the whole sweep (difficult in the MHz range). The classic Howland topology has a known sensitivity to matching and is found in bioimpedance applications — with specialized design guidelines. [25]
- **V-source (constant voltage):** Very common in impedance cytometry: an AC voltage is applied and the differential current is measured; this is also described in the tutorial review (typical setup: applied multi-frequency voltage, measured differential current). [26] It is an easier generator for MHz, but the actual current/field depends on Z and requires current limiting/monitoring.

Sweep strategies:

- **Single-tone stepped-sine sweep:** best SNR per frequency, but slow for very low f (mHz); here we do not target mHz, so it is feasible.
- **Multisine / multi-tone (simultaneous tones):** reduces time and allows real-time/in-line spectroscopy, as also proposed in the report (“multi-tone excitation”). For embedded EIS, modern multiband-multisine strategies exist (especially for time reduction). [27]
- **Chirp (linear/logarithmic):** useful for fast scanning, but requires careful deconvolution and control of non-linearities.
- **PRBS/white noise:** useful for system ID, but more difficult to calibrate for absolute metrology.

Example waveforms (conceptual):

- Sine sweep: $\sin(2\pi f_k t)$ with f_k in logarithmic steps.

- Multisine: with phase selection ϕ_i to reduce crest factor (important for headroom).
- Burst-mode: grouping frequencies into bursts alternating with quiet windows for Vm-like mode (reducing interference).

Maximum allowable Vpp/Ipp (cell viability):

There is no “single value”; it depends on electrode spacing, conductivity, and assay. However, as an engineering envelope based on the literature:

- 10–100 mV (tissue cultures/3D) as an order of magnitude, [8]
- 0.1–0.5 V as a probable order in microfluidics, [8]
- and examples of 0.1–5 V in cytometry papers/tables. [9]

The product specification must define: 1. a default low level, 2. an upper limit per mode, 3. field/current control, and 4. logs/alarms.

4.4.3. Design notes for the impedance AFE (sense + demod)

2-electrode vs. 4-electrode sense:

The 4-electrode (Kelvin) measurement is a classic tool for “eliminating” the electrode/medium interface term from the measured Z (at least to first order), and is reported as a solution to polarization effects. [8]

Lock-in / IQ demodulation:

For each tone f_0 , the measured voltage or current is demodulated into I/Q via synchronous multiplication and low-pass filtering. This is central to achieving SNR in a noisy electrochemical environment, and is explicitly proposed in the report (“lock-in (analogue or digital IQ)”).

Integrated solutions vs. discrete for EIS:

- An integrated AFE such as the AD5940/AD5941 targets bio-impedance/electrochemical measurement, but states an

impedance frequency range up to approximately 200 kHz. [28] This suits low-f ECIS, not 10–30 MHz cytometry.

- For the MHz range, a separate HF path is needed (DDS/PLL source + wideband TIA/sense + I/Q demod or direct RF sampling). Example: a DDS source such as the AD9833 gives output up to 12.5 MHz (suitable as a baseline for a 10 MHz target). [29] For gain/phase measurement and analogue conversion to baseband, RF phase/gain detectors can be used (e.g. the AD8302 with an input range from LF to GHz), although this changes the calibration model relative to a classic lock-in. [30]
- If a Howland current source is chosen for I-excitation, it must be taken into account that the output impedance degrades at high f due to stray capacitances, so practical use in the MHz range requires special architectures or a switch to V-excitation. [31]

4.5. Digitization, DAQ, and signal processing

4.5.1. ADC resolution, ENOB, and trade-offs with SNR/bandwidth

The basic theoretical-SNR equation for an ideal ADC is $\text{SNR} \approx 6.02 \cdot N + 1.76 \text{ dB}$, used to connect bits to dynamic range/quantization noise. [32] In practice, we care about the ENOB (effective number of bits), not the nominal N. [33]

Below is a “chart-table” (theoretical SNR vs. bits) as a quick guide to the trade-off:

Bits (N)	16	18	20	22	24
SNR (dB, ideal)	98	110	122	134	146

[34]

Interpretation for V_m-like mode:

If we want to digitize $\sim\mu\text{V}$ signals in an environment where the front-end may see offsets/common-mode signal, a 24-bit $\Delta\Sigma$ AFE with PGA is very practical because it transfers much of the noise/ENOB difficulty to the analogue domain, close to the sensor. For example, the ADS1299 provides 24-bit resolution, programmable gain, and is documented as designed so that no additional buffer is needed between the electrodes and the ADC, with datasheet-documented noise/CMRR. [19]

Interpretation for impedance mode:

- If IQ demodulation is performed and digitization occurs at baseband (low frequency), the sample rate is not MHz but kS/s; however, the accuracy in magnitude/phase depends on: crest factor, averaging, ADC noise, and front-end linearity.
- If direct RF sampling is used (digital lock-in with sampling on the carrier), MSPS/tens of MSPS are required, along with strict anti-aliasing — high cost/power.

4.5.2. Sample rate per channel and multiplexing

Vm-like mode (0.01–300 Hz): - Adequate f_s per channel: 1 kS/s (or 500 S/s under lower requirements), so as to leave margin for digital filters and avoid aliasing from noise/EMI. - If N_{ch} is large, simultaneous sampling is preferred, or a very fast MUX with correct settle time — but in ultra-low-frequency applications, timing synchronization is less critical than drift/stability.

The report explicitly notes the need for clocking/triggers/time alignment, especially when hybrid data streams are desired.

Impedance cytometry mode: - The literature shows throughput from hundreds to thousands of cells/s, and that high focusing/flow rates impose requirements on “electrical sampling.” [35] - In published datasets/tables, frequencies (e.g. 500 kHz & 10 MHz) and amplitudes appear, aimed at extracting peak/phase features from the temporal trace of each cell. [9]

In practice, if the signal to be demodulated has a bandwidth of e.g. 10–50 kHz (depending on flow speed), a baseband fs on the order of 200 kS/s per “baseband channel” is realistic for correctly capturing peaks without excessive interpolation (exact number: not specified, depends on microchannel geometry).

Anti-aliasing: - For Vm-like: an analogue LP near 300–500 Hz before the ADC, plus digital decimation. - For impedance baseband: LP per I/Q channel (e.g. 1–10× the expected event bandwidth). - For direct RF sampling: band-pass + high-order anti-alias (usually undesirable in a first-generation product).

4.5.3. Digital processing and feature extraction

For impedance fingerprints, the processing flow matches the report: IQ demod/lock-in → fitting to an equivalent circuit / Cole-Cole / multi-shell dielectric models → feature vector → classifier (ML).

The impedance-cytometry literature emphasizes that frequency dependence “maps” sub-cellular regions and that the data is often analysed as magnitude/phase in scatter plots or as fitted parameters. [6]

At the product level, a quality (QC) layer is required: flags for drift, “electrode health,” baseline validation, and self-test via internal known impedances/loopback.

4.6. Safety, isolation, and reliability

4.6.1. Patient-like safety mindset: leakage, isolation, short-circuit protection

Although the use-case may be in-vitro, the “patient-like safety mindset” requirement means that we design as if there were a possibility of patient connection or human handling in a wet environment.

Leakage currents (referencing IEC 60601-1 philosophy):

In a summary of IEC 60601-1 3rd edition requirements, indicative patient leakage-current limits are given:

- Type B/BF: 100 μ A (normal condition), 500 μ A (single fault)
- Type CF: 10 μ A (normal condition), 50 μ A (single fault) [36]

This gives a practical engineering target: even if the product is not certified as ME equipment, the design must keep leakage $\ll$ 10–100 μ A via an isolation barrier and current limiting.

Galvanic isolation:

- An isolation barrier between the “wet-end” (electrodes/AFE) and the host (USB/PC), with reinforced digital isolators and medical-grade isolated power. An example of a digital isolator is the ISO774x (quad-channel), with high isolation ratings. [37]
- For power supply, there are commercial DC/DC series that declare compliance with IEC/EN 60601-1 and $2\times$ MOPP for medical systems. [38]
- On the analogue side, protection against faults/short-circuit: current limiting on the excitation path and sense path, a watchdog that cuts excitation if excessive current is detected or if the ground/reference is “lost.”

Series resistors and input protection:

Even in ECG design references, series resistors are used as patient protection so that fault currents remain below safe thresholds. [39] In your in-vitro system, the same logic translates to: “if an electrode pad short-circuits or liquid spills onto a connector, the instrument must not be able to conduct dangerous current.”

4.6.2. ESD/EMC

As a robustness specification, IEC 61000-4-2 severity Level 4 (8 kV contact / 15 kV air) is often “sufficient” for general-purpose

devices, and there is documented practice in I/O protection applications. [23]

For low-level signals, the EMC strategy must combine: shielding, a proper reference plane, guard rings on high-impedance nodes, and ferrites/filters on the power supply.

4.7. Proposed final specifications and implementations

4.7.1. Mermaid block diagram: AFE + excitation + DAQ + isolation

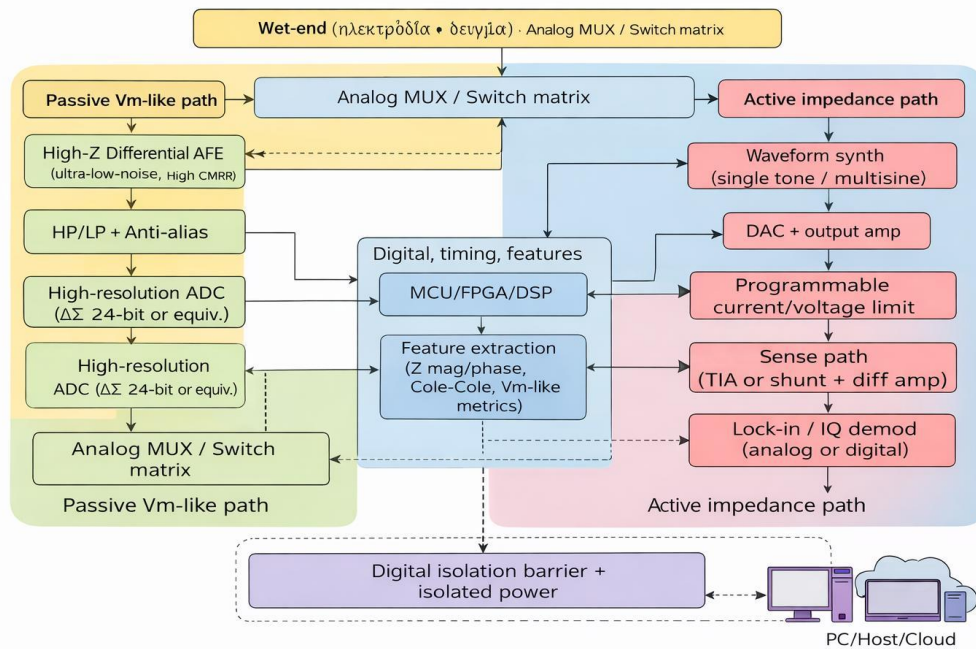


Figure 4.1: Hybrid AFE for Passive Vm-like + Active Impedance (EIS/ECIS) with isolation.

4.7.2. Mermaid signal flow and mode switching

Figure 4.2: Hybrid measurement controller (Passive Vm-like + Impedance mode).

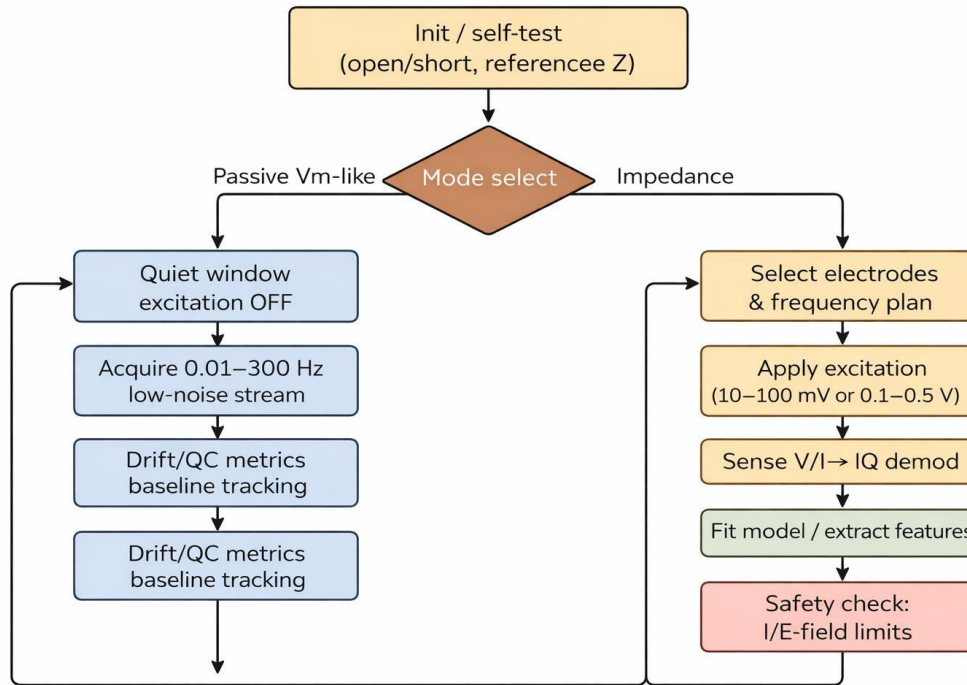


Figure 4.2: Hybrid measurement controller.

4.7.3. Component-level choices (indicative, not exclusive)

Passive AFE/ADC (Vm-like): - 24-bit $\Delta\Sigma$ biopotential AFE: ADS1299 (8-ch) as a reference for low input-referred noise, high CMRR, and high input impedance. [19] - Alternative for large offsets/fast recovery: an ECG-style AFE philosophy that rejects offsets up to ± 300 mV (useful as a design pattern also in vitro, due to electrochemical offsets). [21]

Low-frequency EIS (ECIS-like): - AD5940/AD5941: strong for electrochemical/bio-impedance up to ~ 200 kHz (hence good for low-frequency ECIS, not for 10 MHz). [28]

EIS / cytometry MHz-range: - DDS source up to ~ 10 – 12.5 MHz: AD9833 as a baseline generator for up to 12.5 MHz. [29] - Analogue phase/gain measurement (alternative lock-in implementation): AD8302 (LF to GHz) as a building block, with phase/gain output at baseband. [30] - Current source (if I-excitation is chosen): Howland current-pump design guidelines and known limitations at HF. [31]

Isolation/ESD: - Digital isolator: ISO774x (quad-channel) as an example of robust isolation. [37] - ESD design targets: IEC 61000-4-2 Level 4, as summarized in protection applications. [23] - In the Greek standardization context, the existence of ELOT standards (EN 60601-1, etc.) documents that adoption of the IEC/EN 60601 series is active in the market/regulatory environment (although the full text is commercial). [40]

4.7.4. Table II-2: Proposed final specifications (baseline) and rationale

Category	Proposed baseline spec (range)	Rationale / references
Nch	1–128 (typical start: 8–32, scalable tiles)	The requirement is given as open; the report states an unspecified concept and the need for scaling.
V _m -like amplitude	1–100 μ V (alternatively 10–500 μ V)	Extracellular recordings typically report tens–hundreds of μ V, while non-excitable signals may be much lower. [41]
V _m -like bandwidth	0.01–300 Hz (HP configurable 0.01–0.05 Hz)	Compatible with slow phenomena; ECG references give 0.05–100 Hz as a classic range. [5]
V _m -like input impedance	≥ 1 G Ω	Typical high-Z AFEs such as the ADS1299 state ~ 1000 M Ω . [19]
CMRR	≥ 100 dB @ 50/60 Hz (target 110–120 dB)	Preserves SNR under common-mode; ADS1299 shows 110–120 dB. [19]
Input-referred noise (V _m -like)	Target ≤ 1 μ V _{pp} (0.01–70 Hz) or	Close to state-of-practice for low-noise AFEs. [42]

Category	Proposed baseline spec (range) equivalent in $\mu\text{V}_{\text{rms}}/\text{ENBW}$	Rationale / references
DC offset tolerance	$\pm 300\text{ mV}$ (as a robust design target)	ECG AFEs are designed to reject offsets of $\pm 300\text{ mV}$. [21]
EIS frequency range	1 kHz–10 MHz (proposed extension: 0.1–30 MHz for cytometry)	Cytometry phenotyping bands are often 0.1–30 MHz. [6]
Excitation amplitude (default)	ECIS: 10–100 mV, cytometry: 0.1–0.5 V (with programmable ceiling)	Orders of magnitude for living matter and microfluidics. [43]
Excitation type	V-excitation for MHz, I-excitation for low f/ECIS (or hybrid)	ECIS reports 1 μA AC; cytometry review describes applied voltage + measured current. [44]
Impedance measurement precision targets	(not specified) e.g. $\leq 0.5\%$ magnitude / $\leq 0.5^\circ$ phase	—
ADC (Vm-like)	24-bit $\Delta\Sigma$, fs/ch 0.5–2 kS/s	Low-frequency signal, requiring large dynamic range/low noise. [45]
ADC (impedance baseband)	16–18 bit, fs/ch 50–500 kS/s (per event bandwidth)	Cytometry requires sampling of the temporal trace; the exact fs depends on flow/geometry. [35]
Isolation & leakage targets	Patient leakage design target $\leq 10\text{--}$	IEC 60601-1 summary of leakage limits. [36]

Category	Proposed baseline spec (range)	Rationale / references
	100 μ A (per class), reinforced isolation barrier	
ESD immunity target	IEC 61000-4-2 Level 4 (8 kV contact / 15 kV air)	Common target in system-level ESD. [23]

Note: fields marked as “not specified” must be locked down by the biological endpoint and the assay (adherent vs. suspension, classification vs. monitoring target, throughput, etc.)

4.7.5. Explicit unspecified parameters that must be “locked down”

1. Exact Nch and channel topology (MEA mapping vs. a few cytometry zones).
2. Biological definition of the “Vm-like” observable (what exactly is considered the target signal: field potential, open-circuit drift, etc.).
3. Target SNR / minimum detectable signal per mode (in μ Vrms and in $|Z|$ /phase).
4. Assay constraints: adherent monolayers vs. suspension in flow, target throughput (cells/s), and allowable excitation ($V_{pp}/I_{pp}/E$ -field ceilings).
5. Calibration model (Cole-Cole/ECM vs. multi-shell dielectric fitting) and repeatability/traceability requirements.

The structure of the uploaded report (in particular the recommendation for lock-in/multi-tone, the choice of interdigitated or 4-terminal microelectrodes, and the model layer) already functions as a correct “backbone”; this chapter translates that backbone into concrete electrical specs, architectures, and implementation choices.

4.7.6. Selected baseline implementation (design freeze) and technical rationale

The two-mode hybrid architecture (Passive V_m-like + Active impedance) is locked in, along with the common electrode interface via an analogue MUX/switch matrix and the time-multiplexed strategy (quiet windows + EIS bursts). The following remain configurable: the exact choice of AFE/ADC ICs, the full frequency extension (e.g. up to 30 MHz), and whether the packaging allows 4-electrode Kelvin sensing in a microchannel.

Baseline architecture:

- **V_m-like mode:** passive differential recording with a low bandwidth of 0.01–300 Hz, with design drivers of high input impedance, high CMRR, and control of DC offsets/drift (offset cancellation + baseline/QC tracking). These requirements are classic in biopotential amplifiers, where high input impedance and high common-mode rejection are required.
- **Impedance mode:** multi-frequency EIS/ECIS/impedance cytometry with synchronous IQ/lock-in demodulation to compute the impedance spectrum and parametric features (equivalent circuit/ Cole-Cole), so that the output is a feature vector (rather than a one-dimensional index).

Electrode selection:

As the baseline configuration, an IDE sensing zone is chosen for ECIS/adhesion monitoring on adherent samples, with a reference electrode and guard ring to reduce leakage/parasitics and improve stability. For impedance cytometry, the baseline implementation provides for a second microchannel region with 4-electrode (Kelvin) measurement capability when packaging allows it, so as to limit polarization errors and increase repeatability in liquid samples.

Excitation and demodulation selection:

- For MHz-range cytometry, V-excitation is selected as the baseline due to practical implementation/stability at high frequencies. I-excitation is retained as an alternative only for low-frequency ECIS, where stable, controlled current excitation can be metrologically useful.
- Demodulation is implemented with IQ/lock-in, for high SNR in an electrochemical environment and to support averaging without loss of phase.

Channel scaling. As the initial prototype scale, $N_{ch} = 8\text{--}32$ is defined (tile-based AFE+MUX), with a clear expansion path to 64–128 via repeated blocks. This choice balances throughput against routing/DAQ complexity in the first iteration.

Mode-interaction strategy (interference control). Operation is defined as time-multiplexed: “quiet windows” for V_m -like recording alternate with short EIS bursts (sweep or multi-tone), so as to avoid excitation coupling into the low-frequency passive channel and to maintain a stable baseline.

Safety & isolation baseline:

A “patient-like safety mindset” is adopted: an isolation barrier (transformer/optical/capacitive) and a leakage-control/grounding strategy, aimed at limiting leakage currents and avoiding microshock-type risks.

Chapter 5 - Market Analysis

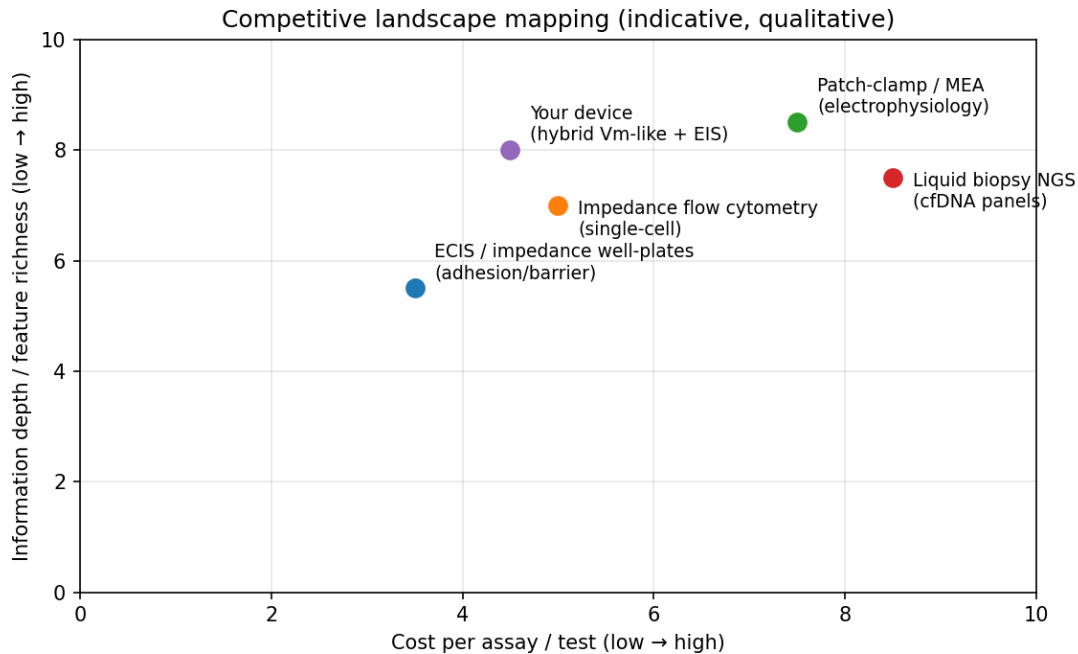


Figure 5.1: Competitive landscape mapping (indicative, qualitative).

Buyers of devices/platforms for liquid biopsy in the EU tend to be split into two procurement “logics”: (a) centralized testing services (send-out) with per-test billing, and (b) an installed base (instrument + kits/consumables, often with reagent-rental/financing). [1]

The oncology market is dominated by ctDNA/cfDNA panel tests (e.g. ~\$3,500 self-pay for FoundationOne Liquid CDx, ~\$5,000 Medicare reference for Guardant360 LDT), while the NIPT (cfDNA) market has international ranges of ~\$350–\$2,900/test depending on country/payer. [2]

The product from Chapter 4 is a hybrid device combining “Passive Vm-like” + “Active impedance (EIS/ECIS/impedance cytometry),” so it can be positioned as phenotypic/biophysical complementary information (mainly relative to CTC workflows) rather than as a direct substitute for genomic ctDNA tests without extensive clinical documentation. [3]

5.1. Product context and regulatory requirements

In the EU, clinical placement of in-vitro diagnostics requires compliance with the IVDR (Regulation (EU) 2017/746) and CE-IVD marking, with stricter risk-based classification and increased documentation/performance requirements. [4]

For laboratories, the usual basis for technical competence is ISO 15189 (in Greece, via ESYD accreditation). [5]

For a device/IVD manufacturer, ISO 13485 is the basic QMS standard practically required by regulatory compliance and the quality chain. [6]

5.2. Buyer & “Job-to-be-done” (who pays, and why)

- **Research laboratories / core facilities (first RUO target)**
 - Want label-free metrics, repeatability, high-throughput, automation (96-well / microfluidic).
 - Pay CAPEX for the instrument + OPEX for consumables.
- **Pharma/biotech (drug screening, MoA, toxicity, resistance)**
 - Want kinetic readouts (hours→days) on adhesion/barrier/viability, and a “feature vector” instead of a single marker.
- **Hospitals/diagnostics (late-stage, after the assay & clinical validation are locked in)**
 - Do not readily buy a “platform” without clinical validation + workflow + reimbursement logic.

5.3. Value proposition

- **Label-free:** no dyes/antibodies → fewer consumables / less prep variability.
- **Functional electrical “fingerprint”:** $|Z(\omega)|$, $\angle Z(\omega)$, fitted ECM/Cole-Cole parameters + V_m -like low-frequency metrics → a “feature vector” (better for classification).

- **Time-multiplexed modes:** quiet windows for passive + bursts for EIS (reduces interference) — engineering control of the measurement chain.

5.4. Buyer profiles and use cases

Segment	“Size” (proxy)	Decision -makers	Procurement drivers	Budgets	Requirements	Use cases: liquid biopsy + hybrid bioelectric
Private diagnostic centres/c hains (Greece/ EU)	Test volume (NIPT + oncolo gy), SLA/T AT	Molecular-lab scientific director, Quality, CFO, IT/LIS	ROI per test, fast TAT, “menu” expansion , financing models	Greece: unspeci fied; EU typicall y capex aligned with platform prices (see below)	ISO 15189, CE- IVD/I VDR kits	ctDNA/cfDNA panels or targeted mutations; NIPT (cfDNA) IVD solutions [7]. Hybrid: fast cell phenotyping/CTC- like after enrichment, label- free triage before NGS/ddPCR (mainly RUO initially).
Hospitals (public/p rivate)	Oncolo gy flow, procurement committees/ten ders	Lab director, pathologist- oncologist KOL, procurement committee, administration	Clinical usefulness /guidelines, reduction of failed biopsies, integration into pathways	Greece: unspeci fied (tender- driven); “cost/outcome” is often requested	IVDR/ CE, ISO 15189, clinical- performance documentation	ctDNA especially when tissue is insufficient/delayed (ESMO). [3] Hybrid: as an “adjunct” for CTC/cellular status (e.g. monitoring, response dynamics), with mandatory clinical validation before diagnostic claims.

Segment	“Size” (proxy)	Decision -makers	Procurem ent drivers	Budgets	Requir ements	Use cases: liquid biopsy + hybrid bioelectric
Research /universit y laborator ies (Greece/ EU)	Grants, throug hput in experi mental protoc ols	PI, lab manager	RUO flexibility , innovatio n, single- cell characteri zation	Budgets : project- based (unspec ified)	RUO accept able (not a diagno stic claim)	Liquid biopsy R&D (MRD/selected mutations, benchmark). Hybrid: core value in label-free impedance & Vm- like signatures and classifier development. [8]

5.5. Competition and delivery models

A) Liquid biopsy (molecular / NGS cfDNA) — indirect competitor

- Guardant360 CDx: a cash-pay rate of \$5,000 (for the uninsured) is reported by Guardant itself.
- FoundationOne Liquid CDx: self-pay \$3,500 is reported by Foundation Medicine.
- For a reference on Medicare ADLT pricing order of magnitude, MedPAC shows \$3,500 for FoundationOne Liquid CDx during the “new ADLT period.”
- For Signatera (MRD), there is a reference to changes in the CMS fee schedule (e.g. ~\$3.5k→\$3.9k) in payer policy commentary.

Correct framing: “They do not measure the same thing.” Liquid biopsy = genomic/epigenetic signal. In our framework: an electrical/functional phenotype on in-vitro cells/samples.

B) Impedance for cell assays (ECIS / label-free well-plates) — direct competitor in the adherent segment

- Indicatively, the Agilent xCELLigence RTCA SP bundle appears at a distributor at a price of \$88,000.
- There is also a refurbished/used market (e.g. €10k–\$20k listings), useful for showing the CAPEX price band.
- Consumables: ECIS 96-well electrode arrays ~\$654 (OPEX per run).

C) Impedance flow cytometry / microfluidic single-cell — direct competitor in the suspension segment

- There are commercial “impedance flow cytometry” platforms (e.g. Amphasys) as a category/technology baseline. (Prices are often quote-based, so do not invent a price if it is not published.)

5.6. Pricing logic

Pricing as an engineering cost model:

- **CAPEX:** platform (AFE/DAQ + isolation + fixture + calibration kit + software stack)
- **OPEX/assay** = disposable chip/plate + standards + operator time + QC repeats.
- **Unit-economics KPI:** cost/assay and throughput (assays/day)

The goal is to fall below the cost order of magnitude of an NGS liquid-biopsy test (thousands of €/ \$ per test) and to provide functional information quickly.

5.7. Comparative table of competitors, Table II.7

Player	Product/ty pe	Model	Price range (public)	Region
Guardant Health	ctDNA CGP/CDx	Test service	~\$5,000 (Medicar e ref)	US + EU (IVDR) [14]

Player	Product/type	Model	Price range (public)	Region
Foundation Medicine (Roche)	ctDNA CGP	Test service	\$3,500 self-pay	US/international [18]
Roche	PCR ctDNA/cfDNA (EGFR v2)	Instrument+kits	Greece: example reimbursement EGFR €160 (not a kit price)	EU/global [19]
Illumina	NGS + IVD NIPT	Instrument+consumables/financing	~\$49k–\$109k (MiSeq i100, US list)	Global/EU [20]
GRAIL	MCED cfDNA	Test service	\$949 list	Mainly US [21]
Sysmex (Inostics/Onco BEAM)	ddPCR ctDNA	Kits/services	Not published (reimbursed in DE)	EU [22]
NIPD Genetics (VERACITY)	cfDNA NIPT	Test service/lab networking	Cost GR/EU: unspecified	EU [23]
Menarini / CELLSEARCH	CTC platform	Instrument+consumables	Not published	US/EU [24]
Bio-Rad	ddPCR	Instrument+consumables	~\$38k–	Global

Player	Product/type platform	Model es	Price range (public) \$55k (starting)	Region [15]
Agilent xCELLigence	ECIS/imp edance (RUO)	Instrument (RUO)	Not published	Global [25]

5.8. Go-to-market & risks

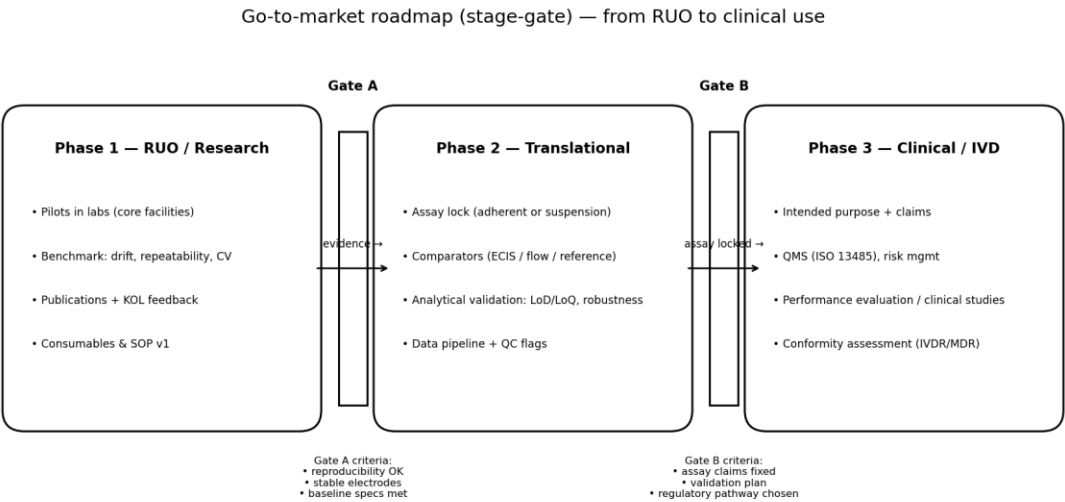


Figure 5.2 — Go-to-market roadmap (stage-gate) from RUO to Translational and Clinical/IVD.

5.8.1. Go-to-Market strategy (why RUO first?)

The main technical uncertainty at present is not the hardware, but the assay lock: adherent vs. suspension, medium conductivity, handling variability, electrode lifetime. So we will first go where iteration is allowed without regulatory weight.

5.8.2. Distribution & adoption (how it actually gets into a lab)

- **Beachhead:** 2–3 lab groups (KOLs) + 1 core facility.
- **Sales model:** “instrument + consumables” (plates/chips) + service.

- **Adoption drivers:** reproducibility, easy calibration, exportable data (CSV/feature vectors), and a “QC dashboard.”

5.8.3. Regulatory & compliance framing (MDR/IVDR)

- If the device is intended for in-vitro diagnosis, then the correct regulation in the EU is the IVDR 2017/746.
- The intended purpose is the legal core: it determines whether it is an IVD, its class, and what evidence is needed.
- For a “medical device” in general, the MDR applies (GSPR Annex I).

5.8.4. Risk analysis

In “Risk → impact → mitigation” form:

(R1) Electrode interface drift / fouling - Impact: degrades repeatability, shifts baseline → false features. - Mitigation: guard/reference, electrode-health metrics, baseline correction, replaceable consumables, saline-phantom routine.

(R2) Assay variability (biological variance) - Impact: non-generalizable feature vector → weak classification. - Mitigation: SOPs, controls (positive/negative), normalization by medium conductivity/temperature, replicate strategy.

(R3) Interference between modes (EIS injection coupling into Vm-like) - Impact: artifacts in the low-frequency stream. - Mitigation: quiet windows + settle time + notch/IQ reject, interlocks.

(R4) Pricing / procurement friction - Impact: labs won’t buy if consumables are expensive or setup is difficult. - Mitigation: RUO pilots, bundle pricing, service plan, clear ROI: assays/day.

(R5) Regulatory “scope creep” - Impact: if we market it as a “diagnostic” too early → we fall under the IVDR and time/cost

balloon. - Mitigation: keep RUO claims until Gate B, then lock in the intended purpose.

(R6) Evidence gap (no comparator truth) - Impact: difficult acceptance by clinicians. - Mitigation: Phase 2 comparators + correlation studies before any clinical claim.

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